A CENTRALIZED SYSTEM FOR REAL-TIME FARM ANIMAL DISEASE REPORTING

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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Abstract

In rural farming situations, diagnosing livestock diseases is made difficult by environmental variation, the imbalance in the available disease datasets, uncertainty regarding the automated predictions, and the morphological differences that exist between various livestock populations. Such factors can hinder the ability of computer-vision systems to generalise and to be practically reliable. The present study designs and assesses a modular pipeline for the analysis of cattle images which integrates object localisation, disease classification, and lesion segmentation.

A development process based on design science was carried out, this involving the harmonisation of the dataset, the construction of the model, iterative refinement, and evaluation. Several annotated datasets were combined for four types of images: cattle, Lumpy Skin Disease, Foot-and-Mouth Disease, and mastitis. For object detection YOLOv8 was employed, a pretrained Vision Transformer (ViT-B/16) was fine-tuned for the purpose of classifying image regions, and a lesion segmentation model based on ResNet50 was developed for the mask head. Where it was documented, augmentation, transfer learning, class-weighted training, oversampling, and early stopping were applied in order to deal with data variation and class imbalance.

The test-set accuracy of the ViT classifier was 90.11%. After refinement, the segmentation mean Intersection-over-Union (mIoU) rose from about 0.298 to 0.4078, representing an absolute gain of 0.1098 over the 100 samples that were evaluated. While these results indicate that a modular cattle-image analysis pipeline is feasible, they do not prove that the system is ready for field deployment. The implementation failed to include a formal out-of-distribution calibration experiment or a controlled evaluation comparing cattle with buffalo; severe imbalance among the minority classes and only moderate segmentation performance still constitute limitations.

The individual's contribution involved preparing and harmonising the datasets, developing the detection, classification, and segmentation stages, training and refining the models, and assessing the recorded experimental results.

Keywords: Computer Vision, Livestock Disease Diagnosis, Vision Transformer, Object Detection, Lesion Segmentation

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LIST OF ABBREVIATIONS

ADRS	Animal Disease Reporting System
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CV	Computer Vision
DAPH	Department of Animal Production and Health
DSR	Design Science Research
FMD	Foot-and-Mouth Disease
IoU	Intersection over Union
LSD	Lumpy Skin Disease
mAP	mean Average Precision
mIoU	mean Intersection-over-Union
OOD	Out-of-Distribution
PLF	Precision Livestock Farming
ROI	Region of Interest
ViT	Vision Transformer
YOLO	You Only Look Once

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1. INTRODUCTION

1.1 Overall Project Context

Livestock production plays a vital role in Sri Lanka's agricultural economy and in the livelihoods of people in rural areas. According to the Department of Animal Production and Health (DAPH), the amount of milk produced in Sri Lanka in 2024 was about 402.69 million litres, with the milk from cows and that from buffaloes being given separate figures in its annual statistical bulletin [1]. These figures offer a current national background for research which aims to improve the timeliness and consistency of livestock health information.

The task of monitoring animal diseases remains one that is carried out by DAPH. In the information it has made public regarding disease control, DAPH outlines national programmes for Foot-and-Mouth Disease (FMD) and other livestock diseases and states that FMD is endemic in Sri Lanka with adverse effects on the dairy sector[2]. Furthermore, DAPH sets out a network of veterinary offices and investigation centres which facilitates passive surveillance, field investigations, diagnostic services, and the reporting of disease status [3].

The livestock disease surveillance system (ADRS) which has been developed as part of this project is a centralised, multi-part system designed to enable earlier disease surveillance based on data. It has four research components: (1) disease diagnosis using computer vision, (2) real-time geospatial clustering and trajectory mapping, (3) seasonal disease forecasting and risk evaluation, and (4) assessment of animal wellness and management. This structure is in accordance with the current design of the project; general reviews of precision livestock farming mention similar applications of AI for animal recognition, abnormality detection, health monitoring, and decision support [4]
.

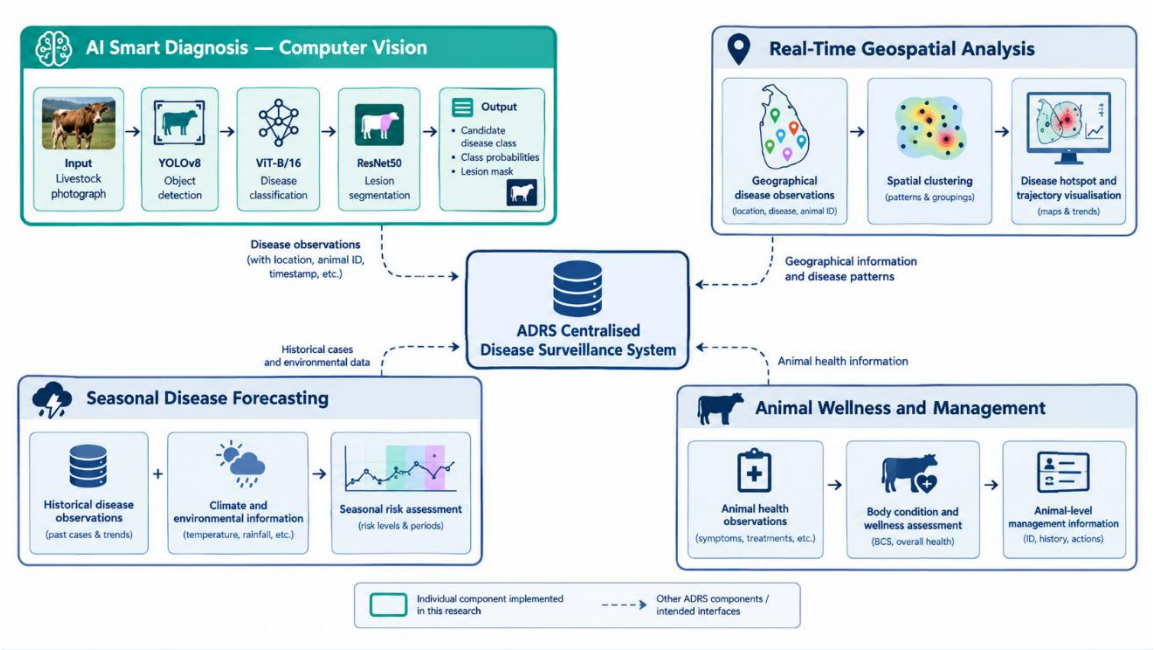


Figure 1 Overall ADRS four-component architecture

The Computer Vision part transforms photographic observations into structured outputs for image analysis that aid the project's other research components. This approach is in agreement with the literature on computer vision in the field of livestock, where image classification, object detection, and segmentation are regarded as complementary tasks for obtaining information at the animal level [5].

1.2 Individual Component Context

The thesis section deals with the diagnosis of cattle diseases using Computer Vision. In the full ADRS architecture, this component is located at the stage where photographic data is turned into structured information relating to disease. According to the proposed system workflow, an image captured by means of a mobile application is sent over for analysis, verification and processing, after which it is converted into diagnostic information that can be used by the other ADRS components[4].

In this particular implementation the pipeline includes YOLOv8 for object detection, Vision Transformer (ViT-B/16) for image-region classification, and a segmentation module based on ResNet50. YOLOv8 is employed for the purpose of object localisation [6], the ViT approach makes use of transformer classification based on

image patches[7] , and ResNet is used to provide a residual-learning backbone [8]. The final segmentation model is a ResNet50-based mask-head version and must not be mistaken for a complete Mask R-CNN implementation [9].

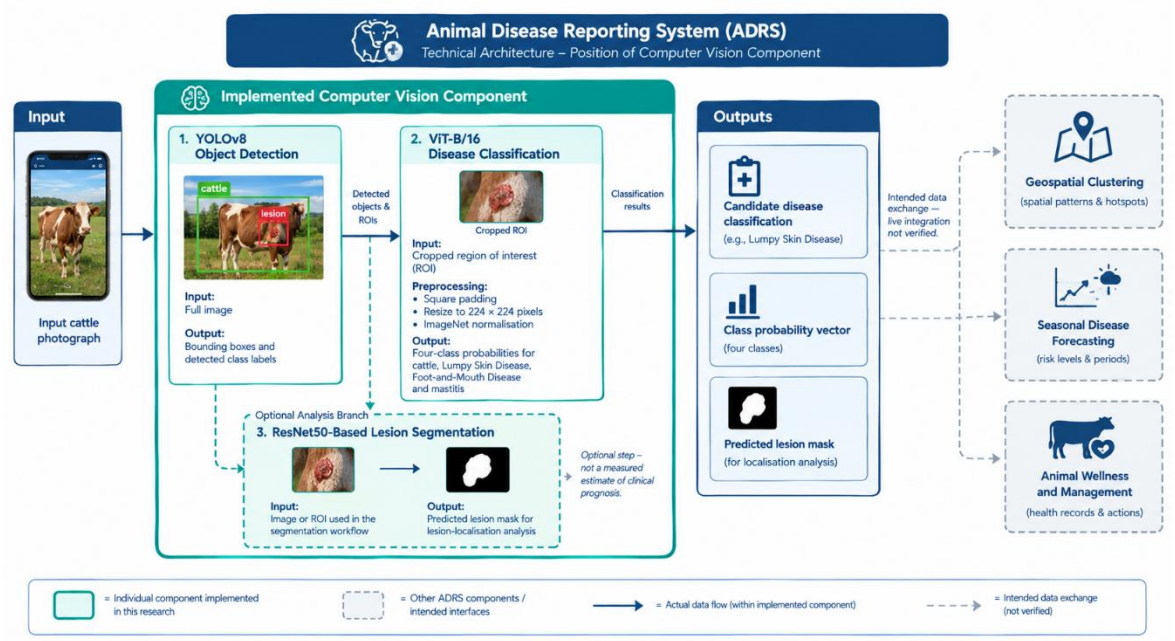


Figure 2 Position of the individual Computer Vision component in ADRS

The component is designed to provide a structured visual signal to the downstream ADRS functions; accurate case descriptions are essential if reliable geospatial summaries and historical surveillance records are to be obtained, and disease forecasts in turn depend on credible historical observations. Just as DAPH's national surveillance strategy combines field reports, disease-status information, and investigation services within its ongoing surveillance and control efforts [3], the computer-vision component is therefore dealing with an upstream data-quality issue by converting raw imagery into consistent candidate observations without considering automated predictions as confirmed veterinary diagnoses.

In the implemented workflow, the detection annotations are used to generate image regions of interest (ROIs), which are square-padded, resized, and passed to the classifier. ROI-based processing is intended to reduce the influence of irrelevant background pixels, although its benefit must be established empirically on the target

dataset rather than assumed [5]. The segmentation stage produces masks that represent lesion pixels and may support a lesion-extent proxy; it does not, by itself, establish clinical disease severity or prognosis.

1.3 Problem Statement

The problem of diagnosing livestock diseases from photographs is one of constrained visual recognition. Photos taken in real-life situations may differ in terms of lighting, the distance from the camera, the angle, the amount of background clutter, the way the animal looks, and the visibility of any lesions. Although deep-learning methods are able to learn discriminative features from images, their performance relies on the representativeness and quality of the training data, and data augmentation is generally employed in order to reduce overfitting and increase the variation during training[5], [10] . A system that has only been evaluated on a single curated subset cannot be taken to perform just as well on images taken under different farm conditions [11].

Another problem is the unequal representation of classes. Since long-tailed datasets have many more examples of common categories than of rare ones, this can cause the learned decision boundary to be biased in favour of the majority classes [12]. The detection dataset compiled for this study included 6,350 cases of cattle, 2,397 cases of lumpy-skin, 946 cases of FMD, and 447 cases of mastitis based on the recorded training and validation annotations. Mastitis thus made up about 4.4% of the cases. Even though weighting and augmentation were used, the report available does not provide class-specific test precision, recall, or F1 scores that would allow us to conclude that the performance for the minority classes was equitable.

The problem in this case is therefore not simply one of training a classifier to achieve high overall accuracy; it is about designing and assessing a modular image-analysis pipeline which is able to localise the relevant areas of the image, classify them according to the available labels, and outline the lesions, while also recording the limitations of the evidence with respect to field robustness, uncertainty, and cross-species generalisation. The outputs from computer vision are regarded as evidence to assist in making decisions, not as a replacement for a veterinary examination or laboratory confirmation [13][14].

1.4 Research Gap

The literature together with the record of the experiment points to four closely related gaps, which are regarded as separate dimensions since improvement in one does not necessarily indicate improvement in the others.

1. **Domain shift**

Computer-vision models generally suppose that the images they are deployed with are similar to those used during training. A discrepancy between the source and the target distributions may lead to a decrease in predictive performance, a problem that has been widely addressed in domain-adaptation research [11]. According to reviews of computer-vision applications in the livestock sector, data diversity and the practical conditions under which the data are acquired remain ongoing problems for real-world deployment[5] [13]. Augmentation can give a model experience of a variety of training transformations[10] , but the fact that augmentation has been used does not prove that the model generalises to actual Sri Lankan farms. The work that was carried out involved photometric and geometric augmentation, though the available evidence does not mention a controlled test carried out across different farms, cameras, or lighting conditions.

2. **Uncertainty estimation**

A softmax output is a set of class probabilities produced by a classification head, but it should not automatically be interpreted as a calibrated probability of correctness. Modern neural networks can be miscalibrated, and post-hoc approaches such as temperature scaling have been studied to improve calibration [15]. Maximum-softmax-probability methods have also been examined as baselines for identifying misclassified and out-of-distribution examples[16]. The implemented ViT produces class-probability outputs, including a recorded example of 91.58% confidence for mastitis, but no calibration curve, expected calibration error, threshold-selection study, or OOD benchmark is reported. Thus, formal uncertainty estimation remains an open gap rather than a completed capability.

3. **Rare disease data scarcity**

The issue of underrepresentation in rare categories is a typical problem in machine learning. While reweighting and resampling are well-known approaches to dealing with long-tailed data distributions, no one method ensures a balanced performance across all datasets [12] . In this study, mastitis was the smallest of the detection classes. For classification, inverse-frequency weighting was applied, and for segmentation, oversampling and class-weight analysis were carried out. Nevertheless, the overall classification accuracy reported is not enough on its own to confirm the quality of the predictions for each of the minority classes; further metrics specific to each class are required.

4. **Interspecies morphological variance**

Writing on livestock vision covers a variety of species and tasks, but individual datasets and studies usually concentrate on specific species, tasks, or phenotypes [5]. Differences in morphology can change the visual characteristics available to a classifier. This becomes important when transferring a model from cattle to buffalo or when using it with other livestock. The current image-classification procedure is centred on cattle and there is no recorded comparison between cattle and buffalo. As a result, interspecies generalisation is regarded as something that must be required in future validation rather than something that has already been achieved.

The main shortage in the existing research is the absence of a continuous and properly evaluated line of evidence that covers localisation, the prediction of disease labels, and lesion delineation, with a genuine assessment of the extent of generalisation. This study provides an implemented three-stage pipeline along with quantitative evidence regarding classification and segmentation, taking care to distinguish the actual measured results from the intended design.

1.5 Aim and Objectives

The aim is to create and empirically assess a modular computer vision pipeline for the analysis of images of diseased cattle, a pipeline which is capable of carrying out localisation, four-class classification, and lesion segmentation, and which also records

the limitations associated with domain shift, uncertainty, performance with rare classes, and variation between species.

The individual work is converted into specific objectives and these are connected to the overall project goals in Table 1. This mapping serves as a tool for keeping track of the work; it does not mean that each group-level aim had been completely realized in the final artifact.

Table 1 Individual research objectives, measurable descriptions, and linkage to group objectives

Objective	Description (Measurable)	Linked Group Objective
IO1 Dataset harmonisation	Merge and harmonise annotated source datasets; document canonical classes and instance counts.	G1 Data collection and annotation
IO2 Object localisation	Train YOLOv8 for cattle/disease-region localisation; report precision, recall and mAP when the evaluation artefact is available.	G2 Input/diagnostic processing
IO3 Disease classification	Fine-tune ViT-B/16 for four labels; report held-out accuracy and class-wise metrics; target aggregate accuracy ≥85%.	G3 Disease classification
IO4 Lesion segmentation	Develop and refine a ResNet50-based mask head; compare baseline and refined mIoU.	G3 Lesion/severity analysis
IO5 Robustness and confidence review	Use documented augmentation and probability output; identify missing calibration, OOD, external-domain and cross-species tests.	G2/G5 Input quality and trustworthy outputs
IO6 Experimental evaluation	Report quantitative results, dataset imbalance and limits; avoid unsupported clinical/generalisation claims.	G6 Ground-truth evaluation

1.6 Research Questions or Hypotheses

This thesis uses research questions rather than null-hypothesis testing because the available experiment summary reports model development and selected performance measures but does not document a controlled statistical comparison across repeated runs.

RQ1: How effectively does the developed YOLOv8 stage localise cattle and disease-related regions in the consolidated annotated dataset, and which evaluation evidence is available to substantiate this?

RQ2: Does the fine-tuned ViT-B/16 classifier meet the project's stated 85% aggregate accuracy target on the reported held-out test set?

RQ3: What effect is observed in the reported segmentation metric after fine-tuning and oversampling compared with the initial baseline?

RQ4: How do class imbalance, augmentation, and class weighting affect the design and interpretation of the developed pipeline, and what class-wise evidence remains unavailable?

RQ5: Do the experiments demonstrate calibrated uncertainty estimation, OOD rejection, real-farm domain robustness, or cross-species generalisation?

RQ6: What are the principal technical and evidential limitations that must be resolved before the pipeline could support a production veterinary workflow?

1.7 Scope, Limitations and Assumptions

In scope

The work involves the construction and harmonisation of a number of annotated datasets, configuring a YOLOv8 detector for the categories relating to cattle and disease, generating crop images from the bounding-box annotations, fine-tuning the ViT-B/16 model for four different labels, developing and improving a ResNet50-based mask-head segmenter, providing quantitative reports on the metrics listed in the implementation summary, and carrying out an analysis of the class imbalance and limitations.

Out of scope

The implementation fails to include laboratory-confirmed diagnoses, a clinical decision rule that has been approved by veterinarians, a validated predictor of recoverability or survivability, the completion of the OOD gate, explainable-AI

attribution maps, a calibrated uncertainty model, formal validation comparing cattle with buffalo, or integration that has been deployed with the other three ADRS components. There is no evidence in the provided experimental summary of full end-to-end latency tests, privacy penetration tests, user studies, or live-farm prospective evaluations.

Assumptions

The text takes it for granted that the provided annotations are appropriate supervisory labels, that the training/validation/test directories were kept separate as stated, that the evaluation samples were not deliberately used for gradient updates, and that the test accuracy and the held-out mIoU that are reported refer to the saved models. Since the summary available does not mention a split grouped by animal identity or by farm identity, it cannot be guaranteed that the samples are independent.

Limitations

The main limitations include the severe imbalance among the classes, the lack of pixel-level segmentation data, the absence of detector score reports, the lack of out-of-domain and cross-species test sets, incomplete records of software versions, and incomplete evidence of production integration. Because of these limitations, the conclusions can only be drawn with respect to the documented datasets and experiments.

1.8 Individual Contribution

The contribution made was in the area of Computer Vision / AI Smart Diagnosis. This involved consolidating a number of object-detection datasets, bringing the labels in line with the four categories that had been implemented, setting up the detection and ROI-processing workflows, fine-tuning the ViT-B/16 classifier, developing a segmentation model based on ResNet50, applying measures to counteract class imbalance, and assessing the documented results of both classification and segmentation.

For the group project, individual responsibilities were assigned regarding geospatial clustering and trajectory mapping, seasonal disease forecasting, and animal wellness/management. This thesis does not claim any of these components as having been implemented individually. The intended interfaces are referred to merely as architectural dependencies; live production integration is not claimed unless additional supporting code and test evidence are included.

The study also included the interpretation of the model's limitations, specifically the fact that the confidence values produced by the ViT were not treated as calibrated uncertainty and that there had been no controlled test comparing cattle with buffalo; this limitation is included as part of the research contribution since it distinguishes performance based on evidence from any proposed future capabilities.

1.9 Thesis Structure

In Chapter 2 the technical and subject-specific literature relevant to livestock computer vision is examined. Chapter 3 sets out the research design, the datasets, the methods of training, the evaluation criteria and the considerations relating to responsible research. Chapter 4 gives an account of the requirements and architecture of the individual component. Chapter 5 describes how the system was implemented, the boundaries of integration, and the testing that was based on evidence. Chapter 6 gives the quantitative results, Chapter 7 interprets these results in relation to the research questions and the existing literature, Chapter 8 discusses the issues relating to commercialization and sustainability, and Chapter 9 contains the conclusions, the achievement of the objectives and the future work.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

Instead, a focused narrative review was carried out rather than a systematically registered one. The sources were found by consulting the publisher's websites, PubMed/PMC, the IEEE/CVF open-access proceedings, OpenReview, Springer, and the official websites of government or organisations. The searches employed a number of term combinations such as "livestock computer vision", "cattle disease deep learning", "lumpy skin disease image classification", "object detection livestock", "vision transformer", "lesion segmentation", "class imbalance", "uncertainty calibration", "out-of-distribution detection", and "domain adaptation". The literature search took place in October 2026.

The selection process focused on peer-reviewed primary studies, review articles that clearly outlined their methods, fundamental computer vision papers, official documentation for the frameworks, and official publications on livestock in Sri Lanka. Sources were chosen because they were relevant to one or more of the four research gaps or to a model that was used in the implementation. Papers containing non-technical promotional content and those whose metadata could not be verified were not included as key evidence. It should be noted that the review does not aim to provide a complete coverage of all papers that have addressed livestock diseases and artificial intelligence; the conclusions are therefore based only on the sources that have been selected.

1. **Domain and Theoretical Foundations**

2. **Precision livestock farming and diagnostic support**

Precision livestock farming uses sensing, data processing, and decision-support methods to support animal management. Recent tertiary review evidence describes AI applications in animal recognition, abnormality detection, health monitoring, and welfare monitoring, while noting that research attention and maturity vary across tasks [4]. Reviews specific to computer vision in livestock likewise report growing use of image classification and other visual methods for obtaining animal-level information

[5]. These findings support the rationale for computer vision as one input modality in an integrated surveillance system, but they do not establish that any individual model is clinically safe or generalises to all field settings.

Object detection

The process of object detection involves determining the class of an object as well as drawing a bounding box around it in an image. In the pipeline that has been implemented, the detector includes four classes: cattle, lumpy skin, foot-and-mouth, and mastitis. A detection model can produce several boxes for each image; the crop-generation stage can then use these coordinates to generate the regions that are to be classified. Ultralytics provides YOLOv8 as a family of models which support both the training and inference phases of object detection [6]. Since there is no formal paper published by Ultralytics regarding YOLOv8, the correct citation for the implementation is its official software documentation. Detection performance is usually described in terms of precision, recall, and mean average precision, and these measures have using calculated using a specific test set with an established IoU threshold.

Vision Transformer classification

The Vision Transformer (ViT) uses a Transformer encoder on a sequence of image patches. Dosovitskiy and others demonstrated that a pure Transformer architecture can achieve strong results in image-recognition tasks when it is pre-trained on a large scale and then applied to new data [7]. In this study, ViT-B/16 was started with parameters that had been pretrained on ImageNet and was modified by substituting its classification head with a four-output layer. The classifier employed processes images that have been square-padded and normalised to a size of 224 × 224 pixels. As a result, the classification stage handles the fine-grained recognition within the crop rather than being responsible for detecting the crop's boundaries.

Residual networks and lesion segmentation

Residual networks include shortcut connections which aid in the training of deeper neural networks by learning residual functions with respect to the inputs to the layers [8]. For its image-feature backbone the implementation used ResNet50 that had been pretrained on ImageNet. A branch based on transposed convolution then generated a pixel-level output. Although the model was said to be inspired by Mask R-CNN, it is not the complete Mask R-CNN architecture, which combines a detection framework with a separate parallel branch for predicting masks in instance segmentation [9]. This distinction in terminology is important: while the implementation does allow for segmentation experiments, it should not be referred to as a canonical Mask R-CNN model.

Dataset imbalance and augmentation

Image augmentation produces modified training examples by means of transformations such as geometric changes, adjustments to colour, mixing, and various other label-preserving procedures. As a broad review points out, augmentation can be used to increase the effective diversity of the training data and to help reduce overfitting in situations where large labelled datasets are not available [10]. It does not generate new ground-truth diversity equivalent to that obtained by collecting new animals, farms, disease stages, or different camera conditions. In the area of long-tailed learning, reweighting and resampling are described as common approaches for reducing the dominance of the majority class; the class-balanced loss literature proposes a weighting scheme based on the effective number of samples [12]. The current implementation employed augmentation and inverse-frequency weighting, but did not use the specific effective-number loss introduced in [12].

Domain shift and uncertainty

Domain shift occurs when the statistical properties of deployment data differ from those represented in the training data. Survey work on medical-image domain adaptation discusses distribution differences and methods for improving transfer between source and target domains [11]. The conceptual lessons are relevant to

livestock images, although medical imaging and farm photography are not interchangeable domains. Calibration is a separate issue: Guo et al. found that modern neural networks can be poorly calibrated and showed temperature scaling as a simple post-processing method [15]. Hendrycks and Gimpel studied maximum softmax probability as a baseline for detecting misclassified and out-of-distribution examples [16]. Neither a softmax score nor a high top-class value is sufficient evidence that a veterinary image lies within the training distribution.

Evaluation metrics

Classification accuracy measures the fraction of examples assigned the correct class. It is intuitive but can conceal poor performance on minority classes when class frequencies are unequal. Precision measures how many predicted positives are correct, while recall measures how many actual positives are retrieved. F1 combines precision and recall by their harmonic mean. Precision-recall curves are particularly informative for imbalanced binary classification because they directly characterise positive-class retrieval and false-positive burden [17]. For segmentation, Intersection over Union (IoU) compares overlap between a predicted mask and its ground-truth mask; mean IoU averages IoU over the evaluated classes or mask categories according to the chosen implementation. Metric definition and averaging conventions must be stated to make results reproducible.

2.2 Related Research

The relevant literature can be divided into three themes. Firstly, reviews have shown that AI and computer vision are capable of supporting a wide variety of livestock monitoring tasks, although they also highlight unequal coverage among these tasks and the need for representative data as well as expert participation [4] [13]. Secondly, disease-specific classifiers show that it is possible to recognise the signs of livestock disease from photographs when using curated datasets; for instance, Saqib et al. assessed a MobileNetV2-based model for lumpy skin disease and achieved 95% accuracy with their healthy-versus-LSD dataset [18]. However, this figure cannot be

directly compared to the 90.11% mentioned in this thesis since the number of classes, the data sources, the splits, the architectures, and the target labels are all different.

Thirdly, foundational work in object detection, Transformers, and segmentation has yielded transferable architectural components [7] [9]. Yet most model papers base their methods on their own benchmark datasets rather than on imagery taken in the Sri Lankan field. The implementation therefore involved the use of established model families, while the original and novel contribution consists of dataset harmonisation and the development and evaluation of components specific to those components rather than a new neural-network architecture.

A major drawback of applied disease classification is its dependence on a single overall metric. Unless per-class precision-and-recall figures, calibration results, or performance data from external domains are available, it is not possible for overall accuracy to show that the model consistently identifies rare or unfamiliar cases. This point is especially significant in this case since mastitis made up only a small part of the combined detection cases and since no formal cross-species split was carried out. The thesis therefore includes the accuracy and mIoU values that are known, notes the missing detector and robustness assessments, and does not make any claim regarding clinical validation.

Table 2 Comparative summary of relevant research and the gaps motivating this study

Study	Method	Findings	Limitation / Gap
Distante et al. [4]	Tertiary review of AI in precision livestock farming.	AI covers recognition, abnormality detection, health/welfare monitoring and decision support.	Broad PLF scope; does not validate this cattle model or its dataset.
Oliveira et al. [5]	Review of deep-learning computer vision in livestock.	Image classification and other visual tasks are used for non-invasive animal-level information.	Species and task coverage varies; field generalisation still requires task-specific evidence.
Xiao et al.[13]	Systematic review of deep learning in veterinary diagnostics.	Reports growing use of DL for veterinary diagnostics and need for diverse data and expert involvement.	Much literature concerns modalities other than farm photographs.
Saqib et al. [18]	MobileNetV2/RMSProp classifier for healthy-vs-LSD images.	Reports 95% accuracy on its binary dataset.	Not directly comparable with a four-label ViT test; different data and protocol.
Dosovitskiy et al. [7]	Vision Transformer image-patch architecture.	Demonstrates Transformer-based image classification after large-scale pretraining.	Foundational benchmark paper, not livestock-specific validation.
Ultralytics [6]	YOLOv8 software and documentation.	Provides detection training and inference implementation.	No formal paper; project-specific detector metrics still required.
He et al. [8] [9]	ResNet and Mask R-CNN architectures.	Foundational residual-learning and instance-segmentation techniques.	The implemented custom mask head is not a full Mask R-CNN.
Guo et al.; Hendrycks and Gimpel [15] [16]	Calibration and OOD-detection methods.	Show why confidence and known/unknown input detection merit separate evaluation.	These methods were not implemented or benchmarked in the current prototype.

2.3 Existing Systems, Products and Techniques

There are several broad types of existing methods. While manual veterinary inspections and laboratory tests can offer a clinical judgement or confirmatory evidence, the availability of these services and the time taken to get results can differ depending on location, the number of cases, and the tests required. DAPH is a nationwide animal-health surveillance and investigation network which includes veterinary offices and investigation centres, and it carries out both passive surveillance and field/laboratory investigations [2] [3]. This system is still essential and an image model should therefore be designed to assist rather than take the place of veterinary decision-making.

Commercial precision-livestock systems make use of sensor devices, cameras, image-processing software, and analytics in order to monitor animals or aspects of their welfare. Reviews cite various applications such as animal recognition, the detection of abnormalities, and monitoring [4] [5]. The feasibility of these systems relies on cost, the existing farm infrastructure, the availability of data, and the way the task is designed. The individual component in question is a software-only image-analysis pipeline which uses standard photographs; its main advantage is that it places a low additional demand on sensors, whereas its main limitation is its sensitivity to image quality and the domain of the dataset.

In the software sector, Ultralytics YOLOv8 provides support for the training and inference of detectors [6], PyTorch together with torchvision is used for training ViT models, and TensorFlow/Keras is used for the segmentation implementation. However, these are development frameworks and not ready-made veterinary diagnostic products. If they are to be incorporated into a safe operational product then it would also be necessary to have validated data lineage, reliable APIs, monitoring, user-facing warnings, permission management, and evaluation by veterinary experts.

2.4 Gap Synthesis and Conceptual Framework

The conceptual framework uses four gaps as evaluation criteria in relation to a three-stage processing pipeline. Since image appearance has an impact on detection, classification, and masking, domain shift affects all the stages. Class imbalance influences both the learning and the evaluation of detection and classification. Species variation compromises the validity of using the learned features on animals other than those in the training domain which is cattle-centred.

The model chain that has been implemented consists of the following steps: input image → detection by YOLOv8 → generation of crops and addition of square padding → determination of class probabilities using ViT-B/16 → mask prediction via ResNet50 when a segmentation is required → production of a structured research output for review. The framework makes no claim regarding a functioning OOD gate or a recoverability engine; these components are part of the original wider project concept and are therefore still considered future development requirements.

The contribution that is expected is thus both technical and based on evidence: it should include an implemented cascade, a quantitative classification result, a comparison between the baseline and the refined segmentation, an explicit explanation of the data imbalance, and a number of clearly identified evaluation gaps which will guide future experimentation.

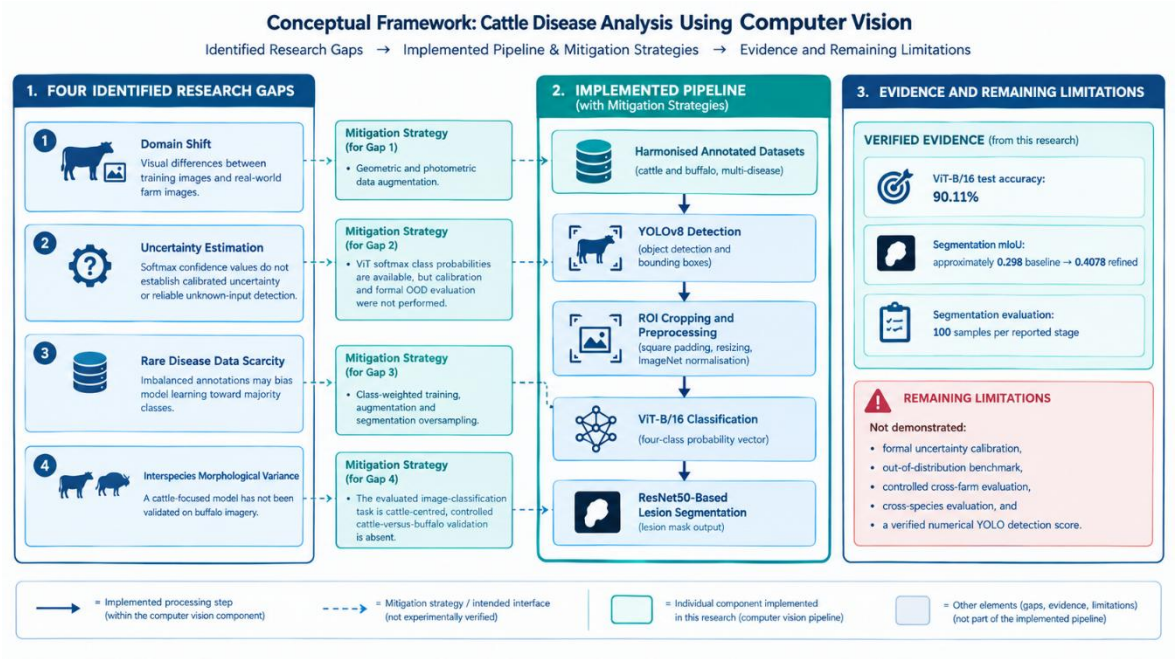


Figure 3 Conceptual framework connecting the identified research gaps to the implemented three-stage pipeline

3. RESEARCH METHODOLOGY

3.1 Research Design

This study follows a design-science-oriented approach. Design Science Research is suitable where the research produces and evaluates an artefact intended to address a practical problem. Peffers et al. describe a process comprising problem identification and motivation, definition of solution objectives, design and development, demonstration, evaluation, and communication [19]. For this project, the artefact is the modular computer-vision pipeline; evaluation is conducted through training histories and held-out metrics documented for the classifier and segmenter.

The work was carried out in an iterative manner rather than as a single, closed experiment. The labels for the dataset were made consistent; classifier crops were generated from the detection annotations; the classifier was trained using transfer learning and with class weighting; the segmenter was trained and then refined; and the results were examined in order to detect signs of overfitting, imbalance, and any remaining gaps. The evidence available is most compelling in the case of the classifier and the segmentation experiments. Although the detector was trained and the best checkpoint was saved, the overall summary of the consolidated experiment does not include a verified mAP/precision/recall score for it.

The study does not involve a prospective veterinary trial or include any human participants; the evaluation described is based on computational methods and datasets. Figure 4 is meant to show the research process, covering the stages from problem definition and dataset construction to model training, evaluation with held-out data, interpretation, and reporting.

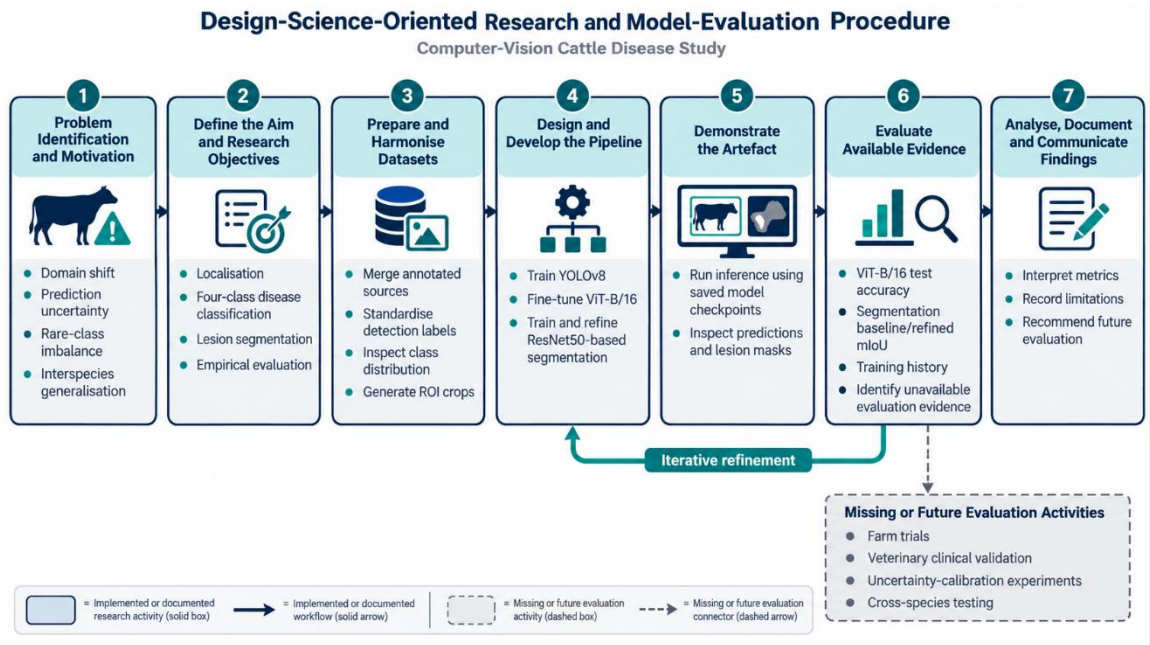


Figure 4 Design-science-oriented research and model-evaluation procedure

3.2 Data, Participants or Study Materials

The dataset used for object detection was created by combining several annotated Roboflow projects relating to lumpy skin disease, FMD, mastitis, and cattle. According to the implementation report, the sources were: atc-gyqoh/lumpy-skin-mdtsn v2; yuossef/lumpy-skin-diesease v6; moaz-ahmen/foot-and-mouth-disease-mfuiw v1; non-fmzye/name1-mb7nq v1; cow-udder/deteksi-mastitis-sapi v3; and kirubel-yemane/cow-and-mastitis-detection v18. The classes from these sources were then converted into four standard YOLO classes: cattle, lumpy_skin, foot_and_mouth, and mastitis.

The total number of instances reported for the combined training and validation annotations was 6,350 cattle, 2,397 cases of lumpy_skin, 946 cases of foot_and_mouth, and 447 cases of mastitis; the training set figures were 5,380, 2,058, 738 and 399 respectively, while the validation set figures were 970, 339, 208 and 48. These are counts of object instances and not necessarily counts of unique images. The consolidated summary does not give the total number of unique images per split; this figure should be obtained from the final dataset manifest if it is available.

In order to carry out the classification, the YOLO annotations were parsed, the normalised coordinates of the boxes were denormalised and the corresponding crop images were extracted. The crops were then organised into train/validation/test folders in a format compatible with ImageFolder, using four class names. A square-padding transformation was applied to maintain the aspect ratio of the crops before resising them to 224 by 224 and carrying out ImageNet normalisation. The precise number of images per split and the number of samples in each class for the ViT dataset are not provided in the summary that was used to prepare this chapter.

The images were used in segmentation training together with COCO-format JSON annotations, most of them coming from FMD and other lesion image collections. The notebook interpreted the image and category annotations and created TensorFlow datasets. The report mentions label and category IDs ranging from 0 to 7 and an eight-class output layer. The refined mIoU was determined using 100 samples that were evaluated. However, the animal-level groupings, the exact sizes of the train/validation/test splits, and the source of the annotations are not included in the available summary.

The primary training hardware mentioned was an NVIDIA GeForce RTX 3050 Laptop GPU; the training of the ViT took about 203 minutes and 52 seconds and was halted after 13 epochs. In the technical summary provided, versions of tools other than the ones specified for the architectures and frameworks were not consistently documented.

3.3 Experimental or Study Procedure

Detection dataset preparation

1) I downloaded annotated datasets in YOLO format from various Roboflow projects. 2) I combined the images and labels into common train, validation, and test directories. 3) The label names were brought into agreement with the four canonical categories. 4) I included background images to serve as negative examples. 5) I set up the dataset YAML file and trained YOLOv8 using augmentation and regularisation settings. 6) The best checkpoint was saved as best.pt for use in inference.

Classification crop preparation and training

For each YOLO label file, the bounding-box coordinates were denormalised based on the size of the corresponding image. 2) The relevant image regions were cropped and the resulting images were placed into folders specific to each class. 3) The cropped images were square-padded and then resized to 224 by 224 pixels. 4) A torchvision ViT-B/16 model was initialised using weights pre-trained on ImageNet and its original head was replaced by a four-class linear layer. 5) Training was carried out using AdamW and cross-entropy loss with the recorded inverse-frequency class weights; mixed precision and image augmentations were also enabled. 6) The validation loss was monitored, early stopping was applied with a patience level of 10, and the best checkpoint was saved. 7) The saved model was evaluated on the test set and an accuracy of 90.11% was recorded.

YOLOv8s Training and Model Selection

For the object-detection phase the YOLOv8s architecture was used in order to detect the cattle and the areas of the images associated with disease. The model was trained on the harmonised dataset which included four categories: cattle, lumpy skin disease, foot-and-mouth disease, and mastitis. During training geometric and photometric augmentation were incorporated in order to introduce more variation into the training images and to decrease the dependence on particular visual conditions.

The training record that is available consists of 111 epochs, and during training the bounding-box regression loss, the classification loss, and the distribution focal loss were recorded for both the training and validation datasets. Furthermore, validation precision, recall, mAP@50, and mAP@50–95 were kept under observation in order to assess detection performance.

The choice of model was guided by the validation mAP@50–95 rather than by the last training epoch. The highest validation mAP@50–95 achieved was at epoch 86. This criterion was adopted since mAP@50–95 assesses detection performance over a number of Intersection-over-Union thresholds and thus offers a more rigorous evaluation of bounding-box localisation than mAP@50 alone.

Progressive improvements in the object-detection metrics were shown during the training process before the validation performance became relatively stable. The learning curves and the evaluation results are given in Section 6.1.

Segmentation training and refinement

1) Loaded the images along with their COCO-format annotations and assigned the image identifiers to the corresponding category annotations. 2) Created the image/mask training samples and resized them to the dimensions required by the model input. 3) Used an ResNet50 that had been pretrained on ImageNet as the backbone together with a convolutional upsampling mask head. 4) Trained an initial model while at first keeping the backbone part frozen. 5) Calculated the class weights and increased the number of samples for the minority classes. 6) Unfroze the upper layers of the ResNet and then carried out further training at a lower learning rate. 7) Compared the mIoU of the baseline model and the refined model across 100 evaluated samples.

Evaluation and record keeping

The classification results were obtained from the saved test evaluation and the segmentation results from the evaluation of the held-out sample. Although the implementation summary referred to the training curves, the loss histories, the confusion matrices and the classification reports, the inference output and the checkpoint files, the consolidated material did not include the full numerical YOLO detector evaluation, the per-class ViT report, the segmentation confidence intervals, and the repeat-seed results. These omissions are regarded as limitations regarding the evidence in the results chapters.

3.4 Measures and Evaluation Metrics

The metrics listed have been chosen so that the evidence corresponds to the relevant objectives; accuracy is given in the case of the four-class ViT test evaluation, and mean IoU is provided for segmentation. Object-detection metrics are designated as the required reporting metrics but cannot be filled in unless the detector evaluation output

is available. For imbalanced categories, precision-recall analysis and the per-class results are better than depending solely on accuracy [17].

Object Detection Evaluation Metrics

The precision, recall, mAP at an IoU threshold of 0.50 (mAP@50), and the mean Average Precision calculated over IoU thresholds from 0.50 to 0.95 (mAP@50–95) were all used to evaluate the YOLOv8s detector.

Precision refers to the ratio of correctly predicted detections, while recall is the ratio of ground-truth objects that have been successfully detected. mAP@50 assesses detection performance by using an IoU threshold of 0.50, and mAP@50–95 gives a more stringent evaluation over a range of localisation thresholds [6].

The bounding-box loss, classification loss, and distribution focal loss were also recorded, and these metrics were looked at in order to evaluate the learning behaviour, convergence, and the differences between training and validation performance.

The most effective checkpoint of the detector was chosen on the basis of the validation mAP@50–95, the maximum value observed being at epoch 86.

Table 3 Evaluation metrics, purpose, and linked research objectives

Metric	Definition	Target / Baseline	Linked Objective
Accuracy	Correct class predictions divided by all test examples.	Reported target ≥85%; observed ViT test accuracy 90.11%.	IO3
Precision / Recall / F1	Precision measures the correctness of positive predictions; recall measures detection of actual positives; F1 is their harmonic mean.	YOLOv8s validation precision and recall approximately 0.80 near epoch 86. Per-class scores and F1 not reported.	IO2 / IO3
mAP@50	Mean Average Precision across detection classes at an IoU threshold of 0.50.	YOLOv8s validation mAP@50 approximately 0.81 near the best checkpoint (epoch 86).	IO2
mAP@50–95	Mean Average Precision averaged across IoU thresholds from 0.50 to 0.95.	Best YOLOv8s validation mAP@50–95 approximately 0.54, at epoch 86.	IO2
YOLO Training / Validation Loss	Bounding-box, classification and distribution focal losses used to assess training convergence and generalisation.	Losses decreased during training; validation localisation losses remained higher than training losses. Evaluated over 111 epochs.	IO2
mIoU	Mean overlap of predicted and ground-truth segmentation masks.	Baseline ≈0.298; refined 0.4078 on 100 samples.	IO4
Calibration / ECE	Measures the difference between predicted confidence and observed correctness.	Not evaluated; future uncertainty-calibration experiments required.	IO5
OOD AUROC / FPR95	Evaluate separation of in-domain and out-of-distribution examples.	No result; OOD model not evidenced.	IO5
Latency / throughput	Time per image and concurrent inference rate.	Not measured in supplied report.	Deployment requirement

3.5 Data Analysis

The analysis is mainly descriptive and comparative. Classification accuracy is given as the final test metric that is available. In order to detect the epoch when the validation loss was lowest and to determine whether a divergence between the training and validation behaviour indicates overfitting, the validation history is looked at. Segmentation refinement is examined in terms of the absolute difference between the baseline and the refined mIoU and also in terms of the relative improvement. Since the summary does not contain the distributions of repeat runs or the paired per-image outputs, a statistical significance test is not carried out.

The number of instances in each class is listed and then expressed as percentages of the total number of detection instances in both the training and validation datasets. This analysis deals with the distribution of the labels and does not assess the fairness of the model or its predictive performance for each class. An appropriate further analysis should include per-class precision, recall, F1 score, confusion matrices, precision-recall curves, and a confidence interval obtained from repeated seeds or from a resampling procedure that has been properly justified.

The qualitative analysis highlights the implementation's strengths and points out potential failure risks, these including the use of multiple dataset sources, class remapping, crop generation, image augmentation, early stopping, and segmentation refinement. Since there was no systematically annotated error set provided in the summary, it is not possible to make any claims regarding the frequency of specific failure modes such as mud occlusion, buffalo confusion, or false positives on unrelated imagery.

3.6 Ethics, Privacy, Security and Responsible AI

The experiments that have been documented make use of annotated image datasets and local annotation files; the implementation summary does not include any information regarding the recruitment of farmers, the collection of identifiable participant data, or an intervention involving animals. Furthermore, it fails to give a formal ethics approval number, animal-use approval, or a source-by-source audit of the data licences. The thesis therefore does not assert that such approvals had been

secured. Before submitting the work, the researcher should check and include the appropriate licences, permissions, institutional ethics decision, and consent records where necessary because primary field images or restricted datasets were used.

For responsible AI practice it is necessary to interpret automated outputs in the context for which they were intended, to communicate any limitations and uncertainties, and to ensure that the stakeholders who are affected remain accountable for decisions. The WHO's guidance on the use of AI in the field of health identifies human autonomy, safety, transparency, accountability, equity and sustainability as important factors in governance [14]. Even though veterinary diagnosis is not the same as human healthcare, these principles offer useful guidance for a decision-support system. The pipeline should be presented as a tool to assist trained veterinarians in their review rather than as an independent diagnostic authority.

Data governance must keep record of the source of the dataset, the licence, the provenance of the annotations, the way the data set was split, and the preprocessing history. When a service is eventually deployed, it should avoid including unnecessary personal or farm-location metadata, regulate access to the images and reports, ensure that the data is protected both during transmission and when stored, and offer documented procedures for data retention and deletion. The research implementation summary fails to show that security testing has been carried out or that production-grade encryption has been implemented; instead, these aspects remain among the requirements for deployment rather than representing controls that have already been put in place.

3.7 Reproducibility and Research Artefacts

The main research artefacts mentioned in the implementation summary are the combined YOLO-format dataset together with its data YAML file, the YOLOv8 checkpoint known as best.pt, the preprocessing scripts which generate crop datasets from the YOLO annotations, the ViT checkpoint called best_vit_model.pth, the Tkinter-based utility for classifier inference, the segmentation training notebook and the model weights, the COCO-format annotation files, and the training and evaluation

logs. The ViT and segmentation phases used PyTorch/torchvision and TensorFlow/Keras respectively.

In order to ensure reproducibility, the artefacts should be versioned together with the identifiers of the dataset source, the software versions, the seeds, the split manifests, the class-to-index mappings, the model configuration, the evaluation scripts, and a README file which explains how to reproduce the metrics. The summary provided refers to certain project paths as local Windows paths; instead, local absolute paths should be replaced by paths given in configuration files or by relative paths. In cases where the artefacts cannot be shared due to licence restrictions, a manifest together with download instructions should be supplied instead.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements were based on the aim of turning cattle photographs into structured image-analysis outputs. As the only implementation evidence available consists mainly of model-training artefacts and not of transcripts from stakeholder interviews, the requirements listed below are regarded as research/technical requirements and must not be understood as the result of a formal user-requirements study.

The general ADRS concept calls for the Computer Vision component to assist with downstream analysis using information on disease type and lesion details. However, the particular prototype in question has more limited capabilities comprising image-based detection, four-class classification, and lesion segmentation. Although the requirements relating to formal uncertainty gating, prognosis, explainability, real-time service latency, and API integration are still regarded as desired future requirements, they are not listed as having been verified.

Table 4 Functional and non-functional requirements for the Computer Vision component

ID	Requirement	Type	Priority / Source
FR1	Accept a cattle photograph and validate input file and dimensions.	Functional	High; image inference utilities documented
FR2	Localise cattle/disease-related regions with YOLOv8.	Functional	High; detector checkpoint recorded
FR3	Classify a crop into one of four implemented labels and return probabilities.	Functional	High; ViT test evaluation and inference utility documented
FR4	Generate a lesion mask from an image or ROI.	Functional	High; segmentation evaluation documented
FR5	Preserve a consistent class-ID mapping across tiers.	Functional	High; mapping consistency needs code-level verification
NFR1	Reach the stated aggregate classification target.	Performance	Partly demonstrated; 90.11% test accuracy
NFR2	Provide calibrated uncertainty and OOD escalation.	Safety/reliability	Not demonstrated; future requirement
NFR3	Support cross-farm and cross-species robustness.	Reliability	Not demonstrated; future evaluation
NFR4	Support secure, auditable, versioned deployment.	Security/maintainability	Requirement only; production controls not tested

4.2 Component Architecture within the Complete Solution

Figure 5 shows the implemented path going from the full image through to the detection boxes, followed by ROI extraction and the four-class ViT probabilities, together with the optional segmentation mask. The figure should clearly show the three implemented stages in distinction from the proposed ADRS interfaces for the geospatial, forecasting, and wellness components. The arrows pointing outwards indicate the intended transfer of data, not proof of actual integration.

The bounding boxes and class identifiers are produced by the detector; the crop preparation procedure then converts the identified areas into the spatial dimensions and applies normalisation that are expected by the classifier. The classifier generates four softmax outputs together with a predicted class, while the segmentation model outputs a mask. It is possible that a future API contract could include the image identifier, the model version, the predicted class, the class probabilities, the bounding boxes, the mask reference, the processing timestamp, and an explicit review status. Since no production API schema or integration test was part of the available evidence, this kind of payload stays at the level of a design proposal.

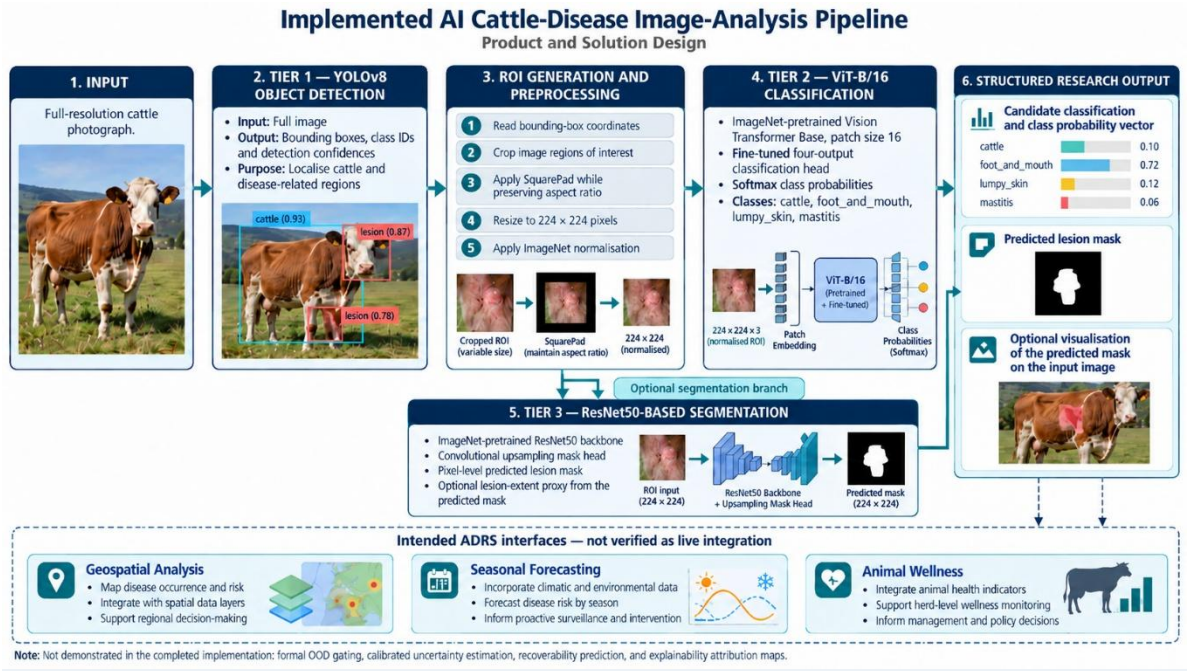


Figure 5 Implemented three-stage Computer Vision architecture and proposed downstream interfaces

4.3 Design Alternatives and Decisions

A single, unified network was not chosen in favour of a cascaded one since the tasks produce different types of output. The output of object detection is spatial regions, that of classification is a discrete label distribution, and that of segmentation is a pixel-level mask. By keeping the various models separate it is possible to train and update each one independently, but this does result in cumulative error: if an object is missed or detected inaccurately then a poor crop will be produced and this will in turn lead to an

incorrect class label. The same problem occurs when the segmentation stage is given a crop that excludes parts of a lesion.

A ViT-B/16 model was chosen and fine-tuned using the pre-trained weights from ImageNet. Vision Transformers represent image structure by means of patch embeddings and self-attention [7]. Convolutional neural networks are still good alternatives and are suitable from a computational point of view when dealing with smaller datasets; since this project did not carry out a controlled comparison between ViT and CNN, the selection of the approach is regarded as a matter of implementation design and not as a claim of demonstrated superiority.

For the segmentation process a ResNet50 backbone together with a task-specific mask head was employed. Although the full Mask R-CNN method is a more detailed instance-segmentation architecture [9], the model that was implemented made use of a lighter custom design for the mask head. The compromise involved reduced implementation complexity at the possible cost of weaker instance localisation and mask quality. The mIoU value of 0.4078 shows that the resulting approach still needs to be further refined.

4.4 Detailed Design

Data and model interfaces

The input for detection is an RGB image and the output is given in the form of bounding-box coordinates, class IDs and confidence values as supplied by the YOLO inference library. The crop generator takes the YOLO labels, converts the relative coordinates into pixel positions, extracts the image patches, square-pads them and then resizes them to 224 by 224 pixels. The classifier gets a normalised tensor as its input and then outputs four class logits before applying the softmax to obtain the probabilities. In the segmentation stage, a resized image or a region of interest is taken as input and a mask estimate is produced.

Class schema

In the merged YOLO dataset the standard detection schema is given as: 0 for cattle, 1 for lumpy_skin, 2 for foot_and_mouth, and 3 for mastitis. The summary of the classifier training shows that the order of the ImageFolder classes is cattle, foot_and_mouth, lumpy_skin, and mastitis, which matches the order of the directories in an alphabetical sequence. Since there is a difference in the ordering between the YOLO and classifier schemas, it is necessary to deal with this explicitly when passing labels between the different stages; the same integer ID must not be assumed to refer to the same class at all stages unless a mapping table is used.

Training and inference behaviour

The training process makes use of the separate training and validation data sets to adjust the model parameters and to keep an eye on its ability to generalise. The test data set that has been set aside is used to give the final reported accuracy. When carrying out inference, the saved checkpoint is used together with the same crop padding, resize operation, and normalisation procedure that were applied during training. This common preprocessing is necessary since alterations to normalisation or to the crop geometry would affect the model inputs. The standalone utility record includes both the predicted class and the probability vector.

Intended diagnostic record

The future diagnostic record must include the image identifier, the input-quality flag where this has been implemented, the model version, the class label, the probabilities, the localisation coordinates, the segmentation output reference, the timestamp, and the review status; it must not state a confirmed clinical diagnosis unless such confirmation has been given by an authorised veterinary procedure. This design for the output promotes auditability and facilitates future interface development.

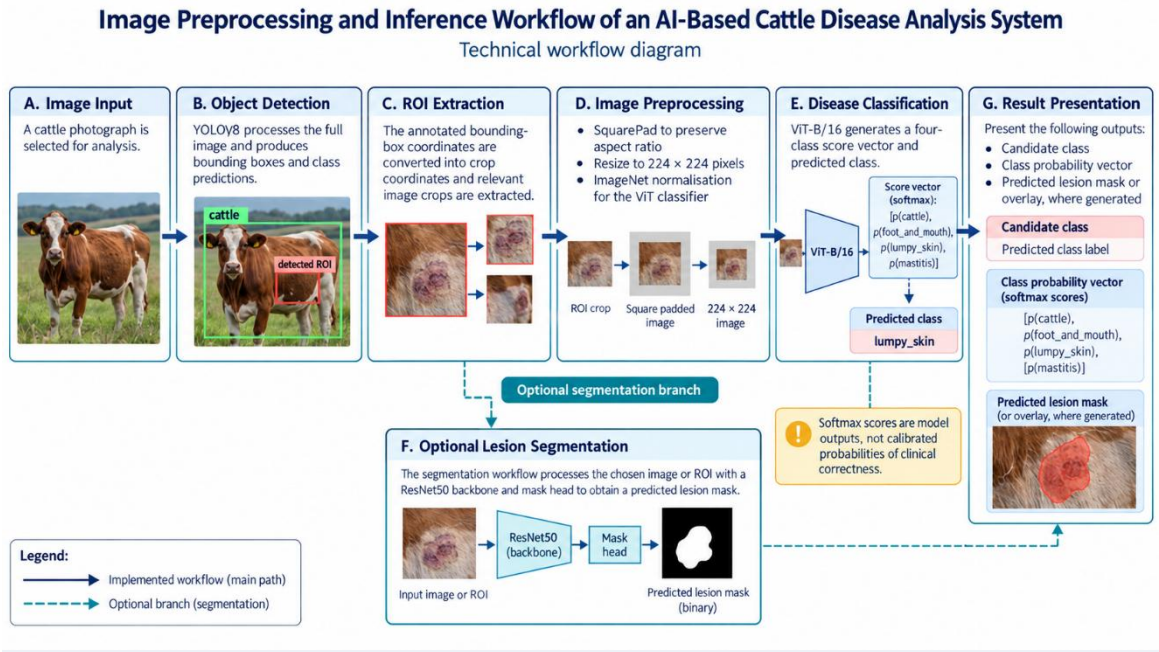


Figure 6 Image preprocessing, model inference, and output-generation workflow

4.5 Standards and Professional Practice

Since this research is carried out in a professional capacity, it is necessary to use datasets that can be traced, provide clear descriptions of the model, ensure that the configuration is reproducible, report the metrics honestly, and make a clear distinction between the experimental output and any clinical conclusions. The licensing and attribution associated with the datasets should be maintained; the class mappings and the split manifests should be kept under version control; and the evaluation must not be chosen or altered following multiple examinations of the test set. The summary given here does not include a formal preregistration or a repeat-seed study, and therefore the results are reported in a descriptive manner.

The application of models within veterinary procedures must be subject to human supervision, along with considerations of safety and accountability. Since the World Health Organization's guidance on AI in health matters emphasizes the importance of keeping ethics and governance at the heart of both the development and use of such technologies [14], it follows that veterinary users should be made aware of the limitations of the models, a procedure for expert review should be maintained, errors

should be monitored across different categories and situations, and validated scores should not be the only basis for deciding on treatment, culling, or outbreak response.

5. IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5 records the documented technology stack. Exact minor versions are included only where provided by the source material; otherwise they are marked as not recorded rather than guessed.

Table 5 Documented development environment and technology stack

Layer / Purpose	Technology	Version	Reason for Selection
Object detection	Ultralytics YOLOv8	Exact package version not recorded; best.pt checkpoint	Single-stage detector and convenient training/inference workflow [6]
Image classification	PyTorch / torchvision ViT-B/16	ImageNet pretrained; 224 × 224 input	Patch-based Transformer transfer learning [7]
Classification training	AdamW, weighted cross-entropy, AMP	Low learning rate; patience 10; stopped at epoch 13	Regularisation, class weighting and compute efficiency
Segmentation backbone	TensorFlow/Keras ResNet50 + mask head	ImageNet pretrained; custom upsampling mask branch	Residual feature extraction and task-specific mask output [8]
Annotation management	Roboflow exports / YOLO labels	Multiple public project versions listed in Section 3.2	Merge and manage bounding-box annotations
Segmentation annotations	COCO-format JSON	Multiple category IDs (0–7) described	Structured image and annotation association [20]
Image processing	OpenCV / Pillow	Exact versions not recorded	Image loading, crop and transform operations
Evaluation	scikit-learn / plotting tools	Exact versions not recorded	Class weighting and analysis of reports/curves
Hardware	NVIDIA GeForce RTX 3050 Laptop GPU	As recorded for ViT training	Available CUDA-capable local training hardware

5.2 Component Implementation

Dataset unification and object detection

Multiple Roboflow object-detection projects were downloaded and merged into shared train, validation, and test directories. Label names were mapped into the four canonical classes. Background examples were included to provide negative training examples. Data augmentation included Mosaic, MixUp, HSV variation, rotation, horizontal flipping, and mixed precision; the configuration summary recorded weight decay of 0.0005 and dropout of 0.1. A best checkpoint was saved as best.pt, and an inference utility was developed to visualise or export detections.

YOLOv8s Object Detection Implementation

The YOLOv8s architecture was used for training the object-detection module on a unified four-class dataset. The model was intended to detect regions of images relating to cattle and disease, and it generated bounding boxes together with class predictions that could be used later for the extraction and classification of the regions of interest.

The training record consists of 111 epochs, and during this process three loss components were kept under observation: the bounding-box loss, the classification loss, and the distribution focal loss. These loss functions were aimed at optimising bounding-box localisation, class prediction, and bounding-box regression.

The training curves recorded indicate a general reduction in training losses together with gains in validation precision, recall, and mean Average Precision; the criterion for selecting the checkpoint was validation mAP@50–95, and epoch 86 was taken as the best performing validation epoch.

The detector forms the first step in the analysis workflow that has been implemented. The bounding-box coordinates as predicted or annotated by the detector can then be used to produce cropped areas for the Vision Transformer classifier. It is necessary, though, to evaluate the model's localisation accuracy on its own, since correct disease classification does not mean that the bounding-box localisation is accurate.

ROI generation and ViT classification

A script used for preprocessing read the YOLO labels, transformed the coordinates by denormalising them, and produced the crop images. Instead of directly stretching the rectangular crops into the square input size, SquarePad maintained the general aspect ratio when resizing. The classifier employed a torchvision ViT-B/16 model with weights pre-trained on ImageNet and had an output layer consisting of four classes. The use of AdamW and weighted cross-entropy was recorded, with the rarest class being assigned a greater weight in the report. The augmentations consisted of horizontal flips, rotations, and colour jitter. Training was carried out using mixed precision and early stopping according to the validation loss; the saved model was assessed on the test set and achieved an accuracy of 90.11%.

Segmentation model

The segmentation annotations were obtained from JSON files in COCO format. The architecture employed a ResNet50 backbone which had global pooling applied and included both a classification branch and a mask branch using transposed convolution ending with a sigmoid output. At first the backbone was frozen. After the initial pass, the upper layers were unfrozen and training was then carried out at a lower learning rate; class weighting and oversampling were applied in order to address the imbalance between categories. The reported mIoU increased from about 0.298 to 0.4078 when 100 samples were evaluated.

Known implementation issues

The summary points out several limitations, including a severe imbalance among the classes, moderate segmentation quality, reliance on a single consumer GPU, the absence of an explicit domain-adaptation method, and the presence of hard-coded Windows paths. The class order used in the detection dataset is different from that stated in the classifier documentation; therefore, the mapping needs to be verified in the final code before any assertion of end-to-end correctness can be made. Moreover, the summary does not indicate that the masks or the model predictions were incorporated into the live ADRS API endpoints.

5.3 Integration with the Other Components

At the model level, the conceptual cascade enables integration since the detector boxes direct the generation of crops, the cropped areas are then classified, and the images or crops can be passed on to segmentation. This is the integrated research workflow referred to in the implementation summary. The material provided, however, does not show live runtime orchestration, a deployed REST API, or the transfer of data into the geospatial, forecasting, and wellness services.

When it comes to a production interface, the data exchange must be explicit and include versioning; the interface has to specify the class schema at each stage, ensure that the class-ID mappings are unambiguous, record the model versions as well as the input preprocessing, and it must be able to tell the difference between a model prediction and a veterinarian-confirmed case. A future integration test should involve passing a known image through the entire cascade and check that the bounding boxes, the label mapping, the probabilities, and the mask dimensions stay consistent across the various models.

5.4 Verification and Software Testing

The testing is carried out in accordance with the evidence given in the implementation summary; the model-training and evaluation results can only be reproduced from the corresponding notebooks if the underlying dataset versions, the package versions, the split manifests, and the checkpoints are kept. The table that follows shows the difference between the model experiments that have been observed and the test types for which no verified record was provided. The fact that something is described as "not evidenced" does not indicate that a failure has taken place; it simply means that the thesis materials do not support a pass claim.

The process of training the object detector was investigated by means of the recorded histories of the loss and the validation metrics. The training logs indicate that convergence was achieved with respect to the main loss functions and demonstrate that validation precision, recall, and mAP all improved during the training process.

The highest validation checkpoint was reached at epoch 86, after which mAP@50–95 stayed fairly stable. This lends support to the idea that the detector had acquired useful object-localisation and classification features within the validation dataset.

The training curves alone do not provide independent verification of the system's performance when applied to a separate external test dataset. It is still necessary to carry out further testing with the help of per-class metrics, confusion matrices, precision–recall curves, and some representative images taken from the field in order to determine how the detector performs for each disease category.

The evaluation of the classifier used a held-out test set and resulted in an accuracy of 90.11%. When it came to the segmentation model, it was evaluated on 100 samples in both its baseline and refined versions. A numerical detection report was not included. Nor were there any formal results relating to runtime latency, production API integration, security penetration testing, usability, or validation by veterinary stakeholders. These absent tests constitute important points for future work and must not be turned into fabricated successful outcomes.

Table 6 Evidence-based verification and test status

Test ID	Description	Expected Result	Actual Result	Status
T01	Dataset merge and label harmonisation	A consistent four-class YOLO dataset.	10,140 train+validation object instances reported; class mapping created.	Documented
T02	YOLOv8 training and inference	Saved detector usable for box prediction; numeric evaluation recorded.	best.pt and inference script recorded; numeric mAP/precision/recall supplied.	Documented
T03	ViT-B/16 classification evaluation	Held-out accuracy ≥85%.	90.11% test-set accuracy.	Target met at aggregate accuracy
T04	ViT early-stopping monitoring	Best checkpoint saved and validation history monitored.	Best validation loss 0.2277 at epoch 3; training stopped after epoch 13.	Documented
T05	Segmentation baseline	Baseline mIoU recorded.	Approximately 0.298 on 100 samples.	Documented
T06	Segmentation refinement	Refined mIoU compared with baseline.	0.4078 on 100 samples; absolute increase 0.1098.	Documented improvement
T07	Uncertainty calibration and OOD rejection	Calibration/OOD metric and escalation behaviour reported.	No calibration curve/ECE or OOD benchmark supplied.	Not evidenced
T08	Cross-species generalisation	Separate cattle/buffalo test results.	No controlled cross-species experiment supplied.	Not evidenced
T09	Live ADRS API integration / latency / security	Verified end-to-end interface and operational metrics.	No production integration or latency/security test report supplied.	Not evidenced
T10	Veterinary stakeholder validation	Documented expert review protocol and findings.	No formal stakeholder evaluation supplied.	Not evidenced

5.5 Deployment and Operation

The artefact that has been implemented is a research prototype consisting of trained model checkpoints and the associated scripts, not a documented production service; the detection checkpoint is stored in a file called best.pt, the ViT checkpoint is saved as best_vit_model.pth, and the standalone classifier utility enables interactive selection of images together with the display of probabilities. Details regarding the exact way of deploying the system and the configuration of the cloud environment or containers are not included in the experimental summary.

When putting the system into operation, the different stages must be wrapped up within a versioned inference interface, make use of configuration-driven paths, verify the dimensions and file formats of the images, record any errors together with the versions of the model, and enforce specific response timeouts. Monitoring should include tracking of image quality, the distribution of classes, changes in confidence levels, false positives, model failures, and the feedback provided by veterinary reviewers. The model should only be retrained with data that is appropriately governed, licensed, and reviewed. These are suggestions regarding deployment and not statements that the actual production controls have already been put into place.

6. RESULTS AND VALIDATION

6.1 Research Results

The chapter includes only the quantitative results that are recorded in the consolidated experiment summary. It divides the material into descriptive dataset statistics, classification performance, segmentation performance, and the issue of missing evidence. Table 6 gives an overview of the main numerical results. These figures are internal findings of the present research and are referred to the tables in the thesis rather than being assigned to external publications.

The dataset used for detection included a total of 10,140 object instances based on the combined training and validation annotations. Of these, cattle made up 6,350 instances (62.6%), lumpy_skin 2,397 (23.6%), foot_and_mouth 946 (9.3%), and mastitis 447 (4.4%). This distribution shows that there is a long-tailed class structure, with mastitis being the smallest category.

The lowest validation loss achieved by the ViT classifier was 0.2277, occurring at epoch 3, when the validation accuracy stood at 92.29%. The training process then continued until epoch 13, when the training accuracy had reached 96.65%, the validation accuracy was 86.13% and the validation loss was 0.5096; early stopping was therefore activated. The final accuracy on the test set was 90.11%.

In the case of segmentation, the initial baseline attained a mIoU of approximately 0.298 across 100 evaluated samples, while the improved model reached a mIoU of 0.4078 over the same number of samples. The absolute difference was 0.1098 and the relative improvement compared to the baseline was about 36.8%. The score is stated as moderate in the implementation summary and must not be referred to as production-grade lesion segmentation.

The YOLOv8s object-detection model was trained using the consolidated cattle and disease-region dataset over 111 recorded epochs. Figure 7 shows the training and validation histories for bounding-box loss, classification loss, distribution focal loss, precision, recall, and mean Average Precision.

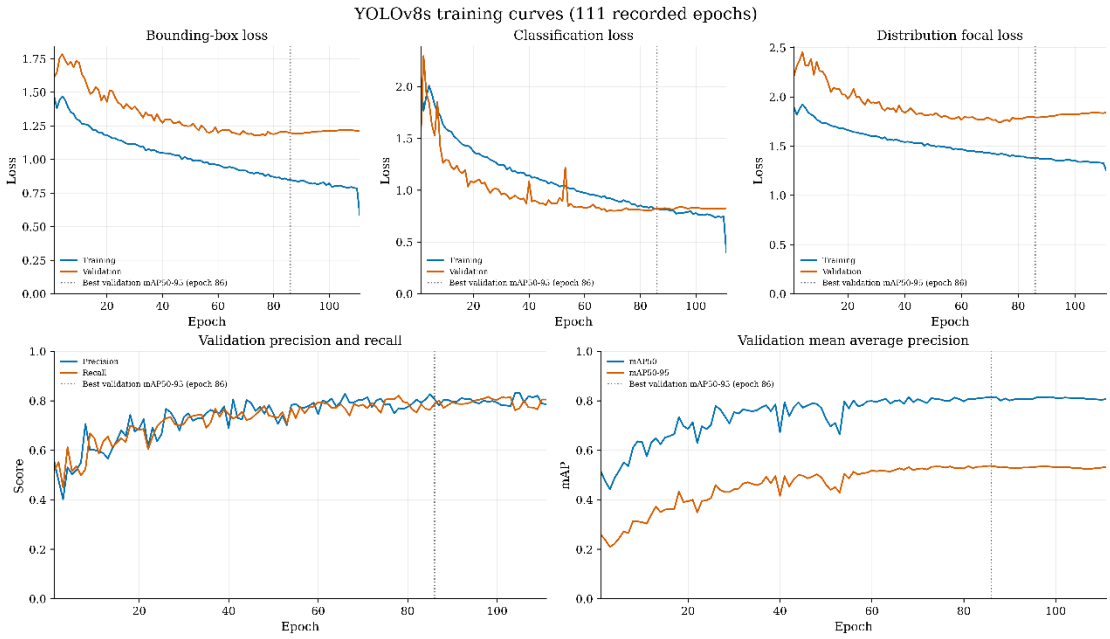


Figure 7 shows the training and validation losses, together with the validation precision and recall and the validation mAP

The training loss for the bounding box decreased from about 1.4 to 1.5 in the early epochs to below 0.8 by the end of the training process. The validation bounding box loss also fell at first but then levelled off to about 1.2. A comparable downward trend was seen in the case of the classification loss, with both the training and validation losses reaching approximately 0.8 by the time the selected checkpoint was reached.

The focal loss associated with the distribution also decreased throughout the training period. Yet the focal loss on the validation set stayed consistently higher than the corresponding training loss, hovering at a relatively stable level of about 1.8 in the later epochs.

The precision and recall for validation rose from about 0.5 in the first phase of training to values close to 0.8 in the later epochs, and the validation mAP@50 increased from about 0.5 to around 0.81, the mAP@50–95 rising from approximately 0.25 to about 0.54.

The highest mAP@50–95 value was achieved at epoch 86, and after that time both mAP values stayed relatively stable without any further sustained improvement beyond the chosen checkpoint.

Table 7 gives a summary of the validation performance which has been estimated from the recorded training curves. The precise figures should be taken from the original Ultralytics training-results file prior to the final submission.

Figure 8 displays the ViT train/validation curves recorded, and Figure 9 shows the comparison between the baseline and refined segmentation mIoU.

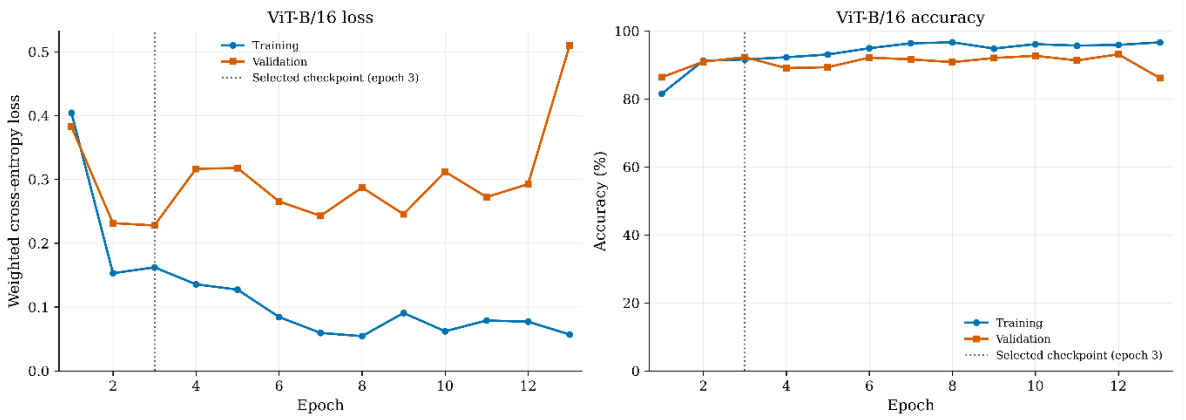


Figure 8 ViT-B/16 training and validation curves

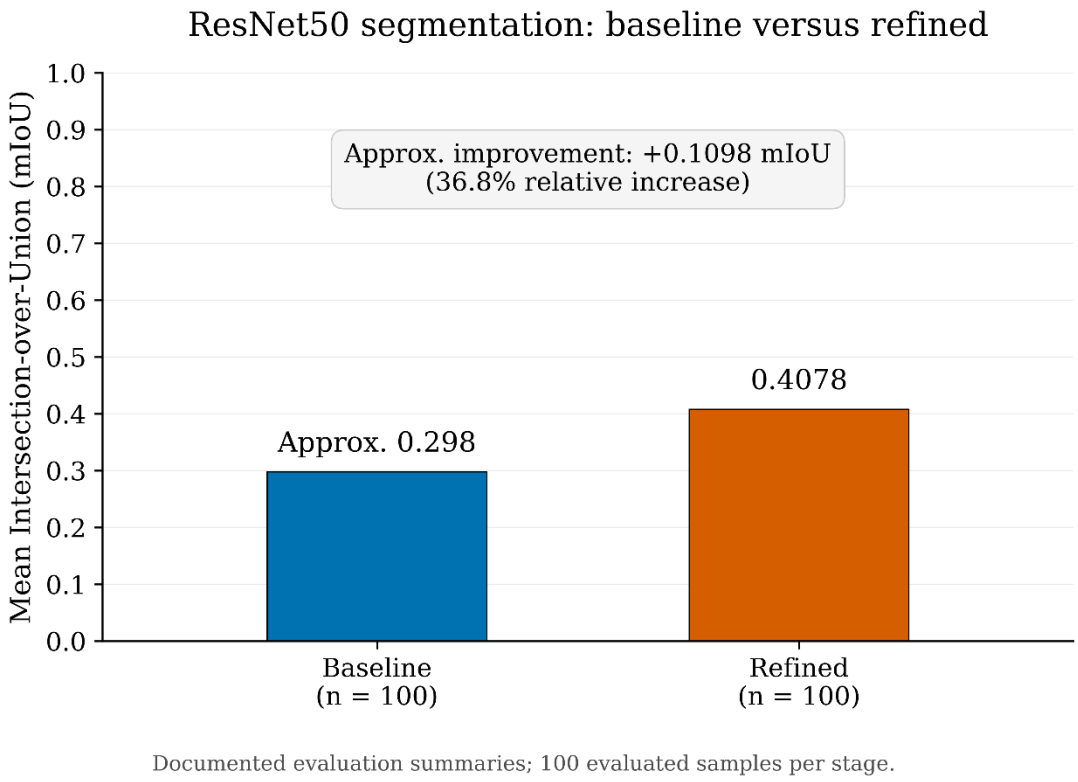


Figure 9 Baseline and refined segmentation mean Intersection-over-Union (mIoU).

Table 7 Quantitative experimental results documented in the implementation summary

Experiment / Condition	Metric 1	Metric 2	Notes
YOLO Object Detection	Validation precision ≈0.80; recall ≈0.80	mAP@50 ≈0.81; mAP@50–95 ≈0.54	Best validation checkpoint at epoch 86 of 111 recorded epochs.
Detection dataset distribution	10,140 combined train+validation instances	Mastitis 447 (4.4%)	No unique-image count or detector score in summary.
ViT best validation epoch	Validation loss 0.2277	Validation accuracy 92.29%	Epoch 3.
ViT final recorded epoch	Training accuracy 96.65%	Validation accuracy 86.13%; validation loss 0.5096	Epoch 13; early stopping triggered.
ViT held-out test set	Accuracy 90.11%	Training duration ≈203 min 52 s	Class-wise metrics not available in the consolidated summary.
Segmentation baseline	mIoU ≈0.298	100 samples	Before refinement.
Segmentation refined	mIoU 0.4078	100 samples	After fine-tuning and oversampling.
Segmentation change	Absolute increase 0.1098	Relative increase ≈36.8%	Moderate remaining mask performance.
OOD / calibration / species test	Not reported	Not reported	No quantitative claims should be made.

6.2 Product and Technical Results

The validation curves for YOLOv8s, as recorded, offer quantitative evidence of the detector's learning performance; at the chosen checkpoint, which is epoch 86, the validation precision and recall were about 0.80, and mAP@50 and mAP@50–95 were about 0.81 and 0.54 respectively.

The fact that mAP@50–95 is lower than mAP@50 shows that the detection performance was worse when assessed using stricter thresholds for bounding-box overlap. Even though the results show that useful localisation is possible within the

validation dataset, they do not prove that the same level of performance is achieved on independent field images.

The numerical results are given as approximate values drawn from the graph of the training curve; the precise figures and the performance for each class should be obtained from the saved validation results before they are finally reported.

The ViT model is said to have attained a test accuracy of 90.11%, which is above the group proposal's aggregate accuracy target of 85% for classification. However, this kind of comparison is limited since the available results do not provide the class-wise F1 scores and the target set out in the proposal related to field conditions, which had not been independently verified by a controlled field test.

The segmentation score became better as a result of the refinement but still stayed at a moderate level of 0.4078 mIoU. Since there was no verified detector metric provided in the summary, it is not possible to assess the detection performance against a specific quantitative standard. Processing latency, peak memory usage, concurrent throughput, energy consumption, failure rate, and availability of the deployed service were not recorded and so are not included as measured product performances.

6.3 User or Stakeholder Validation

The computational validation carried out used the test subset for ViT classification and a separate evaluation based on 100 samples for segmentation. Nothing in the provided materials refers to a formal review of the model's predictions by a veterinary panel, to a prospective clinical study, to a study looking at farmer usability, or to a DAPH stakeholder acceptance exercise. Hence the thesis does not state that validation by clinical stakeholders had been carried out.

For a future stakeholder validation study, a veterinary panel should assess a blinded sample of true positives, false positives, false negatives, poor-quality images, and unfamiliar cases. The protocol should record expert agreement, whether the model output altered triage decisions, the time required to review each case, and the reasons for escalation or disagreement.

6.4 Integration Results

The individual implementation describes a sequential workflow in which the YOLO detections are used to generate crops for the ViT, after which the segmentation model can process the image or the ROI. This creates an integration route between the three models at the design level. However, the provided evidence fails to demonstrate that the trained services were integrated with the geospatial, seasonal forecasting, or animal wellness components of ADRS via tested APIs, nor does it include an end-to-end latency or a full-system accuracy metric.

Hence, the integration status is given as "pipeline design and the individual-stage artefacts are available; production integration has not been demonstrated". This difference ensures that the results obtained at the individual stage are not presented as if they had been measured for the entire end-to-end system.

6.5 Summary of Findings

The key results are that (1) a single four-class detection dataset was put together by combining several annotated datasets; (2) the ViT-B/16 classifier obtained a test accuracy of 90.11%; (3) the improved segmentation model raised the mIoU from about 0.298 to 0.4078 when assessed on 100 samples; (4) a large degree of imbalance in the minority classes still existed; and (5) there was no formal evaluation of domain shift, OOD uncertainty, cross-species transfer, or production integration. These findings provide quantitative answers to RQ2 and RQ3, give a descriptive answer to RQ4, and show that further evidence is needed for RQ1 and RQ5 regarding the detector, robustness, calibration, and integration.

7. DISCUSSION

7.1 Interpretation of Findings

The object-detection phase of the YOLOv8s training shows progressive learning, as the training losses decrease and the validation precision, recall, and mean Average Precision increase, which shows that the detector has acquired useful features for identifying and localising both cattle and disease-related areas in the dataset.

The fact that the training and validation losses are different shows an important limitation with regard to generalisation. While the training bounding-box and distribution focal losses kept on decreasing, the corresponding validation losses stayed considerably higher and did not show any improvement in the later epochs. This discrepancy is in agreement with there being a certain degree of overfitting or a difference in distribution between the training and validation subsets.

The validation mAP curves provide an additional way of interpreting the results. During both the initial and the middle phases of training, both the mAP@50 and the mAP@50–95 increased before reaching a stable level. The highest validation mAP@50–95 was achieved at epoch 86, which indicates that continuing the training past this epoch did not lead to any further improvements in validation detection performance.

The difference between mAP@50 and mAP@50–95 is also considerable. The higher mAP@50 shows that the detector was more successful when using a less strict threshold for bounding-box overlap. The lower mAP@50–95 indicates that accurate localisation continued to be more difficult under stricter overlap thresholds. It is important to note that the downstream ViT classifier depends on relevant image crops; inaccurate or loosely aligned bounding boxes might include unnecessary background information or could exclude important visual details of the lesion.

The results available therefore show that the YOLOv8s detector is useful as an object-localisation stage, although they also point out the need for further testing of localisation precision, minority-class performance, and robustness in real farming

conditions. The results do not, on their own, prove that the domain-shift or interspecies-generalisation gaps have been resolved.

The fact that the ViT achieves a test accuracy of 90.11% shows that the model has learned some useful visual differences among the four labels in its dataset under the test split that was evaluated. While this is a promising result, it does not prove that the model has clinical sensitivity or that it will be robust when applied to new farms and species. Since accuracy gives only a single overall figure, it is still unclear whether the class that was least represented was recognised reliably. As the literature on imbalanced classification warns, the performance on the majority class can dominate the overall results and this is why a class-aware evaluation is recommended [12] , [17] .

The model's generalisation does not appear to have kept on improving as training went on. By the 13th epoch, the training accuracy had reached 96.65%, the validation accuracy was 86.13% and the validation loss had increased to 0.5096. The lowest validation loss had in fact been seen at epoch 3. This kind of divergence is in agreement with the possibility of overfitting, and early stopping offers a reasonable measure of protection, though further runs and a separate test would provide stronger evidence [10].

The mIoU was increased by 0.1098, which amounts to approximately a 36.8% relative improvement compared to the baseline. This indicates that in the experiment in question, unfreezing the upper backbone layers, class balancing, and oversampling were associated with better mask overlap. However, since these modifications were made at the same time and no ablation study has been carried out, it is not possible to determine the individual causal effects of each change. The resulting mIoU of 0.4078 is still moderate and shows that the boundaries, the small lesions, or the category-specific masks might still be imperfect.

The geometric and photometric augmentation only partly dealt with the domain-shift gap. Although augmentation can increase the variation in training [10], no between-domain experiment was carried out. The uncertainty gap was not completely solved since the softmax probabilities were shown without any calibration analysis [15] [16]

. While the scarcity of data for rare categories was recognised and partially overcome by using weighting or oversampling, class-wise results are required to determine whether this approach improved the recall for mastitis. Cross-species generalisation was not included in the controlled evaluation.

7.2 Comparison with Literature, Baselines and Existing Solutions

If the datasets used for classifying diseases are to be compared directly, then the composition of the datasets, the number of classes, the way in which the training and test sets are separated, and the criteria for evaluation must all be the same. Saqib et al. achieved 95% accuracy in their MobileNetV2-based study of lumpy-skin disease using a healthy-versus-LSD dataset [18]. The 90.11% figure obtained in this study is based on a four-label ViT classification task and so involves a different decision problem. The fact that such high accuracies can be attained on carefully curated image datasets does not mean that one model would outperform the other in a situation where they are both deployed together.

The review of computer vision in the context of livestock shows that image classification is frequently used in such studies, although the types of deployment and the representation of different species differ [5]. Similarly, the third study on the use of AI in precision livestock farming also highlights a number of monitoring and decision-support applications as well as pointing out that evidence is still uneven in some areas [4]. The current contribution is in line with the trend shown by this research, but its practical innovation lies in the multi-stage, multi-source implementation together with the reported results from both classification and segmentation rather than in proposing a new backbone architecture.

The performance of the YOLOv8s model was evaluated in terms of precision, recall, mAP@50, and mAP@50–95, all of which are standard indicators of object-detection performance [6]. There is a reason for the difference between mAP@50 and mAP@50–95 since the latter evaluates the performance across a number of Intersection-over-Union thresholds and thus takes into account sensitivity to more stringent localisation requirements.

The fact that the observed mAP@50 is about 0.81 and the mAP@50–95 is about 0.54 shows that the model's performance varies depending on the evaluation criteria employed. It would be wrong to conclude that the model surpasses other livestock-detection methods unless the models are assessed on the same dataset and use the same annotation standards as well as the same evaluation procedures.

This experiment differs from general-purpose object-detection benchmarks in that it makes use of a heterogeneous dataset which includes cattle and various disease-related categories. As a result, the performance figures refer specifically to the annotation distribution that was evaluated, and any comparisons with previously published benchmark results should be regarded as contextual rather than as direct evidence of superiority.

Table 8 provides a qualitative comparison of the implementation with various research approaches and must be understood as such rather than as a benchmark ranking. It is not appropriate to average or rank the performance figures from different datasets without having matched test data and consistent evaluation procedures.

Table 8 Qualitative comparison with selected studies and technical approaches

Aspect	Existing Solution / Baseline	This Work	Difference
Task scope	Many studies focus on a specific recognition task or modality [5] [18] .	Cascaded detection, classification and optional segmentation.	Broader task chain, but end-to-end performance not demonstrated.
Classifier result	Saqib et al. report 95% on a binary healthy/LSD task [18].	ViT-B/16 reports 90.11% on four labels.	Not a direct performance comparison; tasks and test data differ.
Long-tail data	Reweighting and resampling are established options [12].	Augmentation, class weighting and oversampling were used.	Class-wise gains are not available.
Segmentation	Mask R-CNN is an instance-segmentation method [9].	Custom ResNet50-based mask head.	Not full Mask R-CNN; mIoU remains moderate.
Uncertainty	Calibration and OOD methods require dedicated evaluation [15] [16]	Softmax probabilities are shown.	No calibration/OOD experiment reported.
Field and species transfer	Reviews note data diversity and transfer as important issues [5] [11]	Cattle-centred dataset and augmentation.	No external-farm or buffalo test documented.

7.3 Research Contribution

There is evidence from the research that a three-stage design can be carried out by using multi-source annotated images of cattle that have been separately trained for detection, classification, and segmentation; the classification experiment reached a test accuracy of 90.11%, and the segmentation refinement increased the mIoU by 0.1098 as compared with the baseline mentioned. This constitutes useful practical evidence regarding the feasibility of the pipeline, not proof of a new neural-network method.

Another important point is methodological in that the high aggregate accuracy on its own does not eliminate the robustness gaps that have been identified. When

interpreting the numerical results it is necessary to take into account the dataset imbalance as well as the lack of domain-specific, calibration, and interspecies evaluation. That is why the thesis includes both the metrics that have been achieved and a record of the evaluations which are still missing if a trustworthy decision-support system is to be developed.

7.4 Product Contribution

The practical device consists of a harmonised YOLO dataset, a trained YOLOv8 checkpoint, a workflow for cropping ROIs, a fine-tuned ViT-B/16 classifier, an inference tool which gives class probabilities, and a ResNet50-based lesion segmentation model. Because of its modular structure it is possible to update one stage without having to retrain the others, as long as the interfaces and the preprocessing stay compatible.

The item in question is a research prototype and is not being offered as a certified veterinary diagnostic product, as a deployed national surveillance service, or as an autonomous treatment recommendation system. More work needs to be carried out in order to validate the detector's performance, improve the segmentation, include uncertainty-aware escalation, set up data governance, and finish the controlled API and stakeholder testing.

7.5 Limitations and Threats to Validity

Internal validity

The gains achieved through segmentation were the result of several changes carried out at the same time—namely, fine-tuning, class weighting, and oversampling—thus the original data cannot be used to isolate the effect of each individual change. The record of the classifier training indicates the presence of overfitting. The order of the class indices has to be verified against the actual implementation code since the order specified in the documentation for YOLO and ImageFolder is not the same. Because of these points, causal conclusions are limited.

Detection-Stage Limitations

The training record for YOLOv8s does offer some useful validation evidence yet it fails to prove complete external generalisation. The figure available shows aggregate validation metrics and not a detailed breakdown of the performance for each class. As a result, it cannot be determined from these curves whether the detector performs just as well in the case of cattle, lumpy skin disease, foot-and-mouth disease, and mastitis.

It is important to point out that the constructed dataset has a considerable degree of class imbalance. Although the overall detection metrics are strong, this can mask poor performance with respect to the less represented categories. Also, the difference between the training and validation localisation losses shows that there is still a generalisation gap, even though the curves themselves do not indicate what the cause of this gap is.

The evidence available also fails to comprise controlled evaluations with different farms, cameras, lighting conditions, or images of buffalo. It must therefore be regarded as showing the validation performance on the dataset referred to rather than as proving that the system is ready for use in uncontrolled livestock environments.

Construct validity

Test accuracy does not entirely reflect the usefulness of a diagnosis, in particular when there is a class imbalance. The definition of mIoU and the convention used for averaging should be verified against the evaluation code, and the class makeup of the 100 samples that were evaluated should be given. The softmax confidence that is shown should not be taken as a replacement for calibration error or OOD performance. A lesion mask is not the same as a clinically validated disease severity score.

External validity

The data consist of combined public datasets and local annotation files, yet there is no record of a controlled external Sri Lankan farm test or a cross-species test. The results could be influenced by the origin of the images, the camera used, the lighting conditions, the identity of the animal, and the labelling conventions. Furthermore, a

held-out split drawn from the same source distribution does not prove that the system works on previously unseen farms or buffalo populations [5] [11].

Statistical conclusion validity

The report includes only one final accuracy figure and a single baseline or refined mIoU comparison, without giving any information on run-to-run variance, confidence intervals, statistical tests, or the results for each class. The outcomes could be affected by the random initialisation or by the composition of the sample. In future studies it is necessary to use repeated seeds, grouped data splits, and uncertainty intervals for the key metrics.

Mitigation

The study dealt with some of the risks by using held-out evaluation, early stopping, augmentation, class weighting, oversampling, and by providing explicit reports on the limitations. The biggest risks remaining need further testing and more data rather than changes to the language: evaluation using data from external farms, reporting at the class level, calibrated uncertainty, cross-species validation, and integration testing.

7.6 Implications

Veterinarians could make use of a computer-vision pipeline in order to prioritise the review of photographs by showing them the proposed labels and the locations of the lesions. Nevertheless, the pipeline's recommendations should stay advisory until it has been evaluated in a prospective study compared with veterinary assessments and is backed by clinical governance. When an image is uncertain or unfamiliar it should be sent to a human reviewer rather than be treated as a confirmed diagnosis [15] [16][14]
.

Standardised image evidence could be used to supplement field reports provided that the provenance, class definitions, and review status are consistently recorded. Since DAPH is already responsible for disease monitoring, field investigations and laboratory services, the future ADRS component should complement these existing systems rather than avoid them [2] [3].

For researchers the findings indicate that it is necessary to report not only the overall performance but also the distribution of the data, the method of splitting the data, the errors for each class, the versions of the model, and any domain shift. A record of the experiment that is capable of being reproduced is essential when assessing whether future improvements actually represent better generalisation or merely result from a more favourable data split.

8. COMMERCIALISATION AND SUSTAINABILITY

The chapter looks at the potential value and sustainability of the Computer Vision component as a research prototype; since there are no records of a market survey, a willingness-to-pay study, a piloted project with costs, or a formal procurement assessment, the commercial statements are presented as opportunities and requirements for further validation rather than as proven market findings.

8.1 Value Proposition and Target Users

The idea is to turn images into structured candidate disease labels and lesion masks by means of ordinary photographs, possibly enabling a veterinary officer to prioritise cases for examination. The main groups of people who would use it are field veterinary surgeons, veterinary investigation personnel, livestock agencies, and larger dairy farms. As noted in [4] and [5], the term "AI-for-precision-livestock reviews" refers to a wide range of applications for automated animal monitoring and decision support, which does support the relevance of this kind of tool, but it does not prove that there is market demand for ADRS specifically.

The main advantage is that it will assist with workflow by providing earlier detection of a possible lesion, offering more structured case evidence, and ensuring that the way image observations are recorded is consistent. It does not state that veterinary travel, production losses, or mortality have already been reduced; instead, such outcomes would have to be assessed in a prospective pilot study, one that measures time to triage, agreement with veterinary review, the amount of false alerts, and actionability.

8.2 Market and Competitive Position

The main methods involved are manual inspection, reports made in veterinary offices, laboratory confirmation when necessary, and the use of specialised precision-livestock devices and software. As shown by the functions relating to surveillance and disease control published by DAPH, the intended environment already encompasses existing public veterinary services and investigation procedures [2][3]. Although the computer-vision prototype is distinguished by its proposed capability to analyse still images by means of detection, classification, and lesion masking, no procurement process was

carried out, no competitor pricing was obtained, and no head-to-head usability study was conducted.

Its position in the competitive field will depend on having been validated in relation to local breeds and conditions, on being reliable when it comes to minor diseases, on having a low operating cost, on being compatible with existing veterinary work processes, and on having clear controls involving human review. Although the present implementation has promising aggregate classification evidence, it still lacks sufficient validation regarding the detector, the field domain, uncertainty, and clinically.

8.3 Business or Sustainability Model

The different ways in which the service could be deployed would be either through an institutionally funded service provided by a livestock authority, a subscription arrangement with veterinary clinics or cooperatives, or a service based on research and incorporated into an existing reporting platform. Once again, these are merely available options and not the ones that have been chosen or validated as revenue models. Instead, a responsible pilot would begin by working out the cost of inference per image, the amount of effort required for model maintenance, the cost of cloud storage, the requirements relating to data governance, the volume of support needed, and the operating cost of having human veterinary reviews.

To keep the system going, it would be necessary to monitor data drift, periodically examine the errors, add new disease images that have been documented under the relevant permissions, and revalidate the updated checkpoints. The model updates would have to include versioning and rollback procedures in order to avoid an untested model silently replacing a previously validated version. The long-term sustainability of the system relies on the institution taking ownership of the workflow and continuing to have access to the appropriate annotation expertise.

8.4 Scalability, Intellectual Property and Compliance

Because of its three-stage design, the models can be retrained independently, but the overall throughput is limited by the slowest of the stages—that is, the cost of uploading the images, the segmentation latency, the memory requirements, and the service

orchestration. The implementation was trained using an NVIDIA RTX 3050 laptop GPU; neither a load test nor a benchmark involving concurrent inference was reported. It therefore follows that any plans for production scaling should be regarded as unvalidated and should be measured before making any service level assertions.

The software at the end has to retain the attribution and licence terms for Roboflow datasets, the pretrained weights, and the libraries. According to the Ultralytics YOLOv8 documentation, licensing terms and guidelines for citations are provided, covering AGPL-3.0 and enterprise licensing options [6]; the relevant terms must be compared with the model that is intended to be used for distribution and hosting. The permissions for the dataset should be checked separately from the software licences.

Photos can contain personal or farm-related metadata. When the system is deployed it is necessary to specify the allowed access, how long the data will be retained, the procedures for deletion, logging practices, and security measures in a way that conforms to relevant law and to institutional policy. The summary of the prototype does not show that a full legal compliance review, a security audit, or a privacy impact assessment has been carried out. The system should act on only the minimum amount of data, avoid including farm information in public reports, and maintain human responsibility for important decisions.

8.5 Contribution to the Sustainable Development Goals

The system has a reasonable connection with Sustainable Development Goal 2 (Zero Hunger) since the health of livestock affects the productivity and resilience of food systems. The United Nations defines SDG 2 as being concerned with ending hunger, attaining food security and better nutrition, and promoting sustainable agriculture [21] . The way in which this component could make a contribution is indirectly: more organised disease data could lead to timely veterinary reviews and enhance the quality of surveillance information.

This study did not assess any measurable impact on the Sustainable Development Goals. Assertions regarding greater milk production, less disease transmission, higher farmer incomes, or reduced mortality would need to be based on a field study with preestablished outcomes. Possible adverse effects might be false reassurance, false

alarms, unequal performance when applied to minority groups or underrepresented species, and an inappropriate dependence on automated outputs. Sustainability must therefore be linked to validation, human supervision, inclusive data collection, responsible licensing, and transparent monitoring.

9. CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

A modular computer-vision pipeline for the analysis of images of cattle was developed in this research, the pipeline combining YOLOv8 for object detection, ViT-B/16 for classification, and a segmentation component based on ResNet50. The experiments recorded a test accuracy of 90.11% for the ViT model and an improvement in the segmentation mIoU from about 0.298 to 0.4078 when 100 samples were evaluated. The results demonstrate that classification and lesion-mask refinement are technically feasible within the datasets that were evaluated.

The experiments in object detection showed that it is feasible to use YOLOv8s for identifying cattle and areas related to disease in the combined dataset. At the best checkpoint, which was reached at epoch 86, the validation mAP@50 and mAP@50–95 came to about 0.81 and 0.54, respectively. Even though the detector showed improvements in terms of precision, recall, and localisation, the gap between training and validation loss together with the lack of an independent field evaluation mean that further refinement and testing are still needed before the system can be put into practical use.

The evidence does not show that all four of the research gaps have been filled; although augmentation and class weighting were employed in order to deal with visual variation and imbalanced classes, neither an external-farm domain-shift experiment nor a demonstration of performance with balanced classes was reported. Softmax probabilities were generated but were not calibrated as formal estimates of uncertainty. The model was focused on cattle and no controlled comparison between cattle and buffalo was carried out. As a result, the conclusion supported is that this is a promising research prototype with measurable results at the component level and with clear needs for further validation, not a system that has been clinically validated and is therefore ready for autonomous use.

9.2 Achievement of Objectives

Table 9 included as a final objective-traceability summary. The dataset harmonisation has been carried out; model construction has been carried out for the three stages; the ViT accuracy target has been achieved at the aggregate level; improvement has been made to the segmentation although it still remains moderate; formal uncertainty/OOD validation and cross-species generalisation have not been demonstrated; and full production integration is still unverified.

Table 9 Summary of individual objective achievement based on documented evidence

IO1 Dataset harmonisation	Achieved in documented artefact	Four canonical classes and reported train/validation instance counts.
IO2 Object localisation	Implementation completed; quantitative evaluation incomplete	best.pt and inference utility documented; mAP/precision/recall values supplied.
IO3 Disease classification	Partly achieved / aggregate target met	90.11% test accuracy; per-class report and external-domain evidence unavailable.
IO4 Lesion segmentation	Partly achieved	mIoU improved from ≈0.298 to 0.4078; remaining performance moderate.
IO5 Robustness and confidence analysis	Partly addressed	Augmentation and probability outputs exist; no formal calibration, OOD or cross-species validation.
IO6 Experimental evaluation	Partly achieved	Classifier and segmentation metrics reported; detection/system/stakeholder tests missing.

9.3 Individual Research and Product Contributions

The research carried out by the individual includes the data-harmonisation workflow, the implementation of YOLOv8 together with the associated checkpoint, the process for generating ROI crops, a fine-tuned ViT-B/16 classifier which provides probability outputs, the ResNet50-based lesion segmentation implementation, and the classification and segmentation evaluations that are reported. The study sets out a method for combining separate model stages into a modular pipeline for the analysis of livestock images and includes measurable evidence together with the limitations identified in order to assist with further development.

This component is meant to supply the broader ADRS system with structured candidate image evidence. The geospatial analysis, the seasonal disease forecasting, and the wellness/body-condition analysis will continue to be handled as separate groups of components. The thesis at present must not state that their APIs or their full-service deployment have been completed unless further test artefacts are included.

9.4 Future Directions

The evaluation outputs for YOLOv8 should initially be recovered and made publicly available, covering per-class precision, recall, mAP at the specified IoU thresholds, confusion matrices, and a test split that is clearly documented. Future enhancements to the object-detection part should involve evaluating it using an independent test set, carrying out a precision and recall analysis for each class, and looking into the cases of false positive and false negative detections. The difference between the training and validation localisation losses also needs to be investigated further by using stronger regularisation, increasing the diversity of the dataset, and applying targeted augmentation. Further evaluation should assess the impact of the quality of the bounding boxes on the downstream ViT classifier and check whether the detection performance stays consistent across different farms, various camera conditions, different disease stages, and among cattle or buffalo populations.

The ViT should also be evaluated using per-class precision, recall, F1, macro-F1, balanced accuracy, and precision-recall curves, with particular attention given to mastitis. The mapping from class indices to labels must be verified at all stages.

The model's segmentation should be evaluated using a bigger and more varied set of pixel-annotated data, along with class-wise IoU measurements, visual overlays, and multiple repeated runs. An ablation study must independently examine the impact of oversampling, class weighting, and backbone fine-tuning. Stronger segmentation backbones should only be investigated once the evaluation split and the annotations have been verified.

Domain shift should be tested on farms that have been held out, on cameras, on different lighting conditions and on different geographic areas, and the data should be grouped by animal or farm where possible in order to reduce leakage. The calibration

methods such as temperature scaling and OOD methods should be evaluated using known in-domain, poor-quality and novel images [15][16]. For cross-species generalisation a study should be used which compares cattle with buffalos and which includes adequate representation together with veterinary-reviewed labels.

The integration work should define and test the versioned API, the image metadata policy, the way errors are handled, the latency targets, the logging procedures, role-based access, and the veterinary review status. A pilot involving animal-health experts should be approved, subject to ethical oversight, and assessed using operational metrics such as the false alert rate, review time, agreement, and the proportion of cases that are escalated for examination. Production claims must come after the evidence has been produced, not before it.

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GLOSSARY

Annotation: The process of assigning labels, bounding boxes, or pixel-level markings to images so they can be used for supervised machine-learning tasks.

Backbone Network: The primary feature-extraction section of a deep-learning model. In this research, ResNet50 was used as the backbone of the lesion segmentation model.

Bounding Box: A rectangular region defined by coordinates that identifies the position and approximate extent of an object or feature within an image.

Calibration: The degree to which a model's predicted confidence values correspond to its observed prediction accuracy.

Checkpoint: A saved version of a trained model containing learned parameters that can be loaded for further training, evaluation, or inference.

Class Imbalance: A condition in which some categories contain significantly more training examples than others, potentially causing models to favour majority classes.

Confusion Matrix: A table comparing predicted classes against actual classes to show correct classifications and different types of classification errors.

Data Augmentation: A technique for increasing training-data variation by applying transformations such as rotations, horizontal flips, colour adjustments, and other image modifications.

Data Leakage: A situation in which information from validation or test data influences model development or training, potentially producing overly optimistic evaluation results.

Domain Shift: A difference between the data distribution encountered during model training and the distribution encountered during evaluation or deployment.

Early Stopping: A training technique that terminates model training when a monitored validation metric stops improving according to a specified criterion.

Fine-Tuning: The process of adapting a pretrained model to a new task by updating some or all of its learned parameters using task-specific training data.

Ground Truth: Reference annotations or labels used to evaluate model predictions. In this study, ground truth includes image class labels, object bounding boxes, and lesion annotations.

Held-Out Test Set: A portion of the dataset reserved for final model evaluation and not used to update model parameters during training.

Inference: The process of applying an already-trained machine-learning model to input data to generate predictions.

Lesion Mask: A pixel-level representation identifying the region of an image predicted or annotated as affected tissue.

Model Generalisation: The ability of a trained model to produce useful predictions on previously unseen data rather than only on examples encountered during training.

Overfitting: A condition in which a model learns training-data-specific patterns that do not generalise effectively to unseen examples.

Pixel-Level Segmentation: An image-analysis task that assigns a label or mask value to individual pixels to identify the spatial extent of a target region.

Softmax Output: A vector of model-generated class scores normalised to sum to one. These values are not necessarily calibrated probabilities of prediction correctness.

Transfer Learning: A machine-learning technique in which knowledge acquired during pretraining is reused and adapted to a different task or dataset.

APPENDICES

Appendix A: Research Instruments or Extended Results

This appendix provides supplementary information regarding the datasets, model configurations, and experimental results of the Computer Vision component. The research involved three deep-learning models: YOLOv8s for object detection, ViT-B/16 for disease classification, and a ResNet50-based architecture for lesion segmentation. Multiple publicly available datasets were combined and standardised into four categories: cattle, Lumpy Skin Disease, Foot-and-Mouth Disease, and mastitis. The consolidated detection dataset contained 10,140 annotated object instances across the training and validation subsets.

The YOLOv8s model was trained for 111 recorded epochs, achieving its highest validation mAP@50–95 at epoch 86, with an approximate mAP@50 of 0.81 and mAP@50–95 of 0.54. The ViT-B/16 classifier achieved a test accuracy of 90.11%, while the refined segmentation model improved mean Intersection-over-Union from approximately 0.298 to 0.4078 across 100 evaluated samples. These results demonstrate the capabilities of the individual models within the evaluated datasets.

The supporting research artefacts include training notebooks, dataset annotations, preprocessing scripts, model checkpoints, training curves, and evaluation outputs. These materials provide additional evidence of the implementation and evaluation procedures described in Chapters 3, 5, and 6.

Appendix B: Academic Integrity and AI Use Disclosure

Generative Artificial Intelligence tools, particularly ChatGPT by OpenAI, were used to assist with the preparation of this thesis. AI assistance included organising research content, improving academic writing, identifying relevant literature, refining technical explanations, and preparing conceptual diagrams illustrating the proposed system architecture, methodology, and research framework.

The experimental results presented in this thesis originate from the research project's model-training and evaluation activities rather than AI-generated experimental data. AI-assisted diagrams were used for explanatory purposes and do not represent independent experimental findings. Responsibility for verifying technical information, evaluating experimental results, reviewing references, and ensuring the accuracy and originality of the final thesis remains with the author. The use of AI assistance is acknowledged in accordance with the academic integrity requirements of the institution.